UNIVERSIDAD NACIONAL AUTÓNOMA DE HONDURAS

CIUDAD UNIVERSITARIA



INGENIERÍA EN SISTEMAS

REDES DE DATOS II – IS611

SECCIÓN: 1000

Grupo #1: Avance de proyecto

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5 de Noviembre 2026

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INTRODUCCIÓN

El presente informe describe el diseño y configuración de una red de comunicaciones para la empresa Fast Food Factory, con el propósito de conectar sus ocho sedes ubicadas en los países: Honduras, Guatemala, Costa Rica, Panamá y Estados Unidos. Este proyecto plantea una infraestructura que permite compartir información y acceder a servicios desde las diferentes localidades de la empresa.

Para ello, se organiza la red de cada sede según sus áreas de trabajo, esto mediante Vlans y se configura una asignación automática de direcciones IP. La comunicación entre sedes se establece a través de rutas estáticas y de los protocolos de enrutamiento dinámico EIGRP, OSPF y BGP. Además, se incorporan técnicas de traducción de direcciones NAT (Network Address Translation) y PAT (NAT con sobrecarga) para que equipos con una IP interna se comuniquen usando otra dirección visible desde fuera de la red.

A lo largo de este informe, se presentan la topología utilizada, las configuraciones y las pruebas realizadas para evaluar la comunicación entre equipos. Con este trabajo se busca aplicar los conocimientos adquiridos en el curso y comprender como sé integran los distintos componentes de una red que conecta varias sedes.

OBJETIVOS

OBJETIVO GENERAL

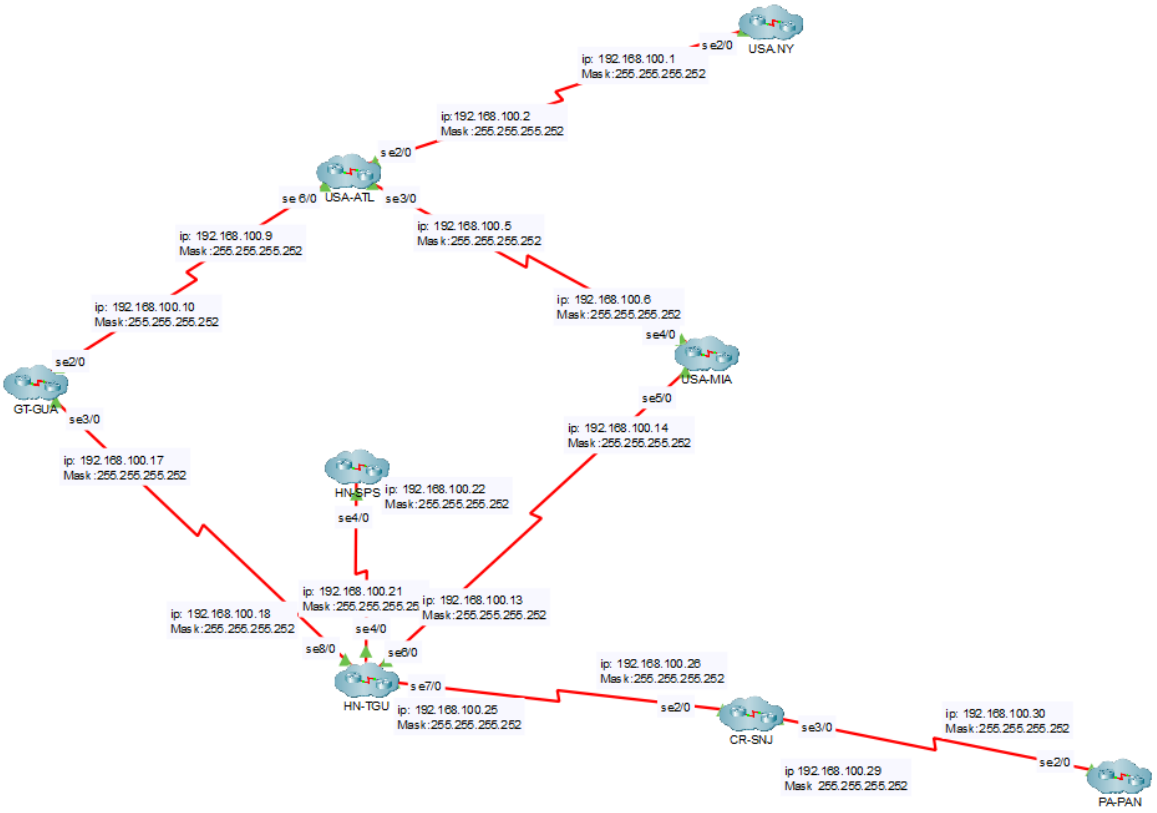
- Diseñar e implementar en Cisco Packet Tracer una red de comunicaciones que interconecte las ocho sedes de Fast Food Factory, permita el intercambio de información y facilite el acceso a los servicios de la empresa.

OBJETIVOS ESPECÍFICOS

- Organizar las redes de cada sede mediante VLAN según las áreas de trabajo y configurar DHCP para la asignación automática de direcciones IP.
- Establecer comunicación entre sedes mediante rutas estáticas y los protocolos EIGRP, OSPF y BGP, incorporando NAT y PAT para la traducción de direcciones según los requerimientos del proyecto.
- Verificar el funcionamiento de la red mediante pruebas de conectividad y acceso a los servicios, documentando las configuraciones realizadas y los resultados obtenidos.

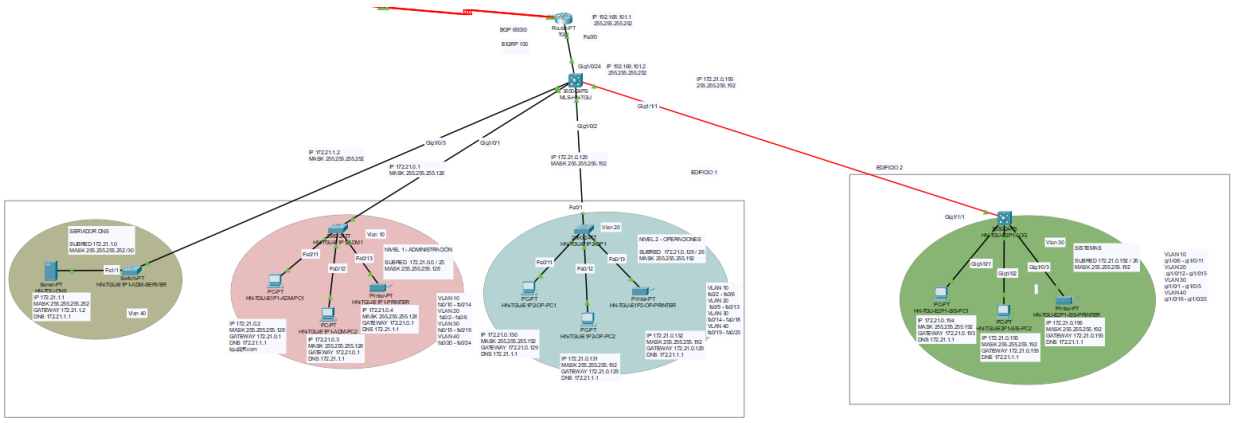
TOPOLOGÍAS USADAS

WAN

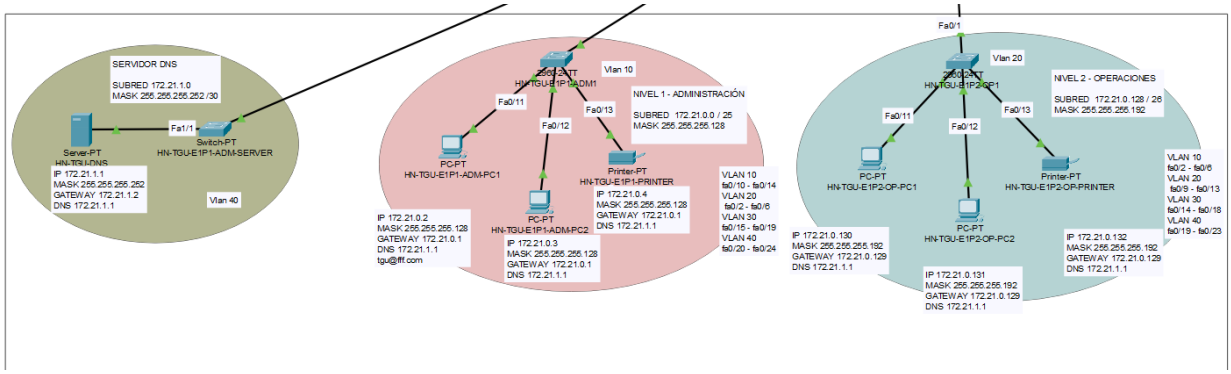


LAN

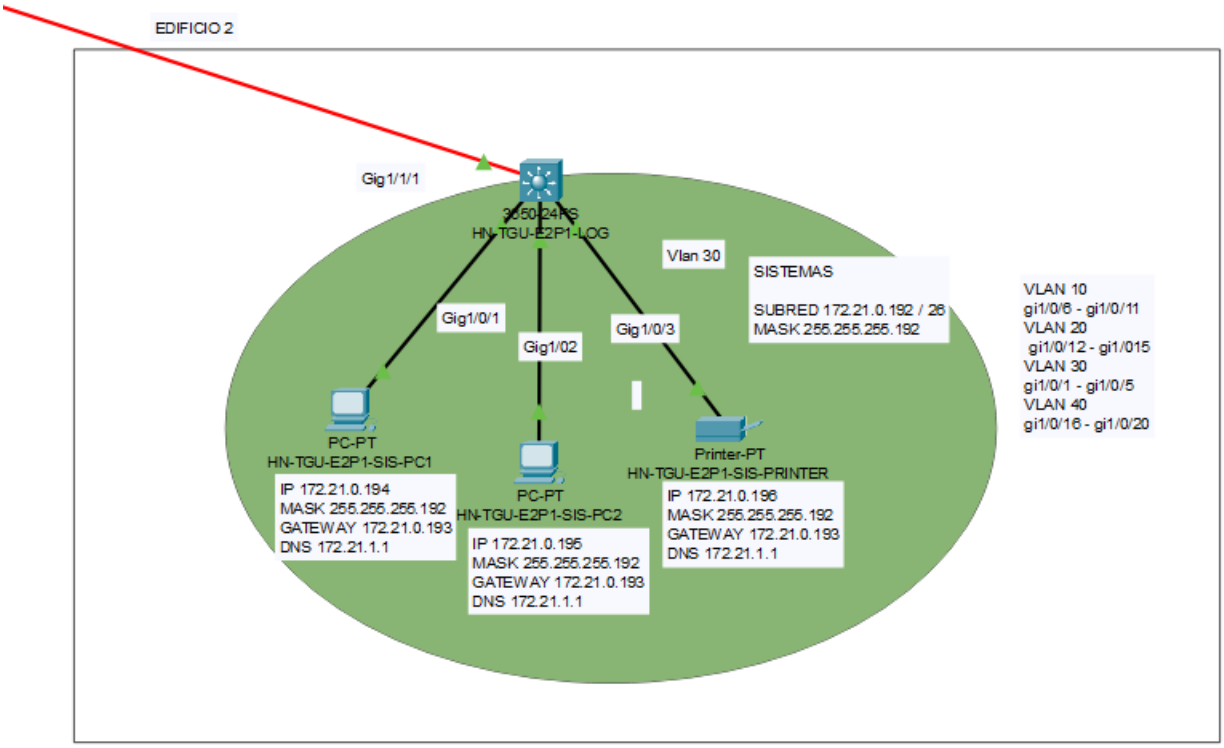
Tegucigalpa (Honduras)



Edificio 1, Tegucigalpa (Honduras)

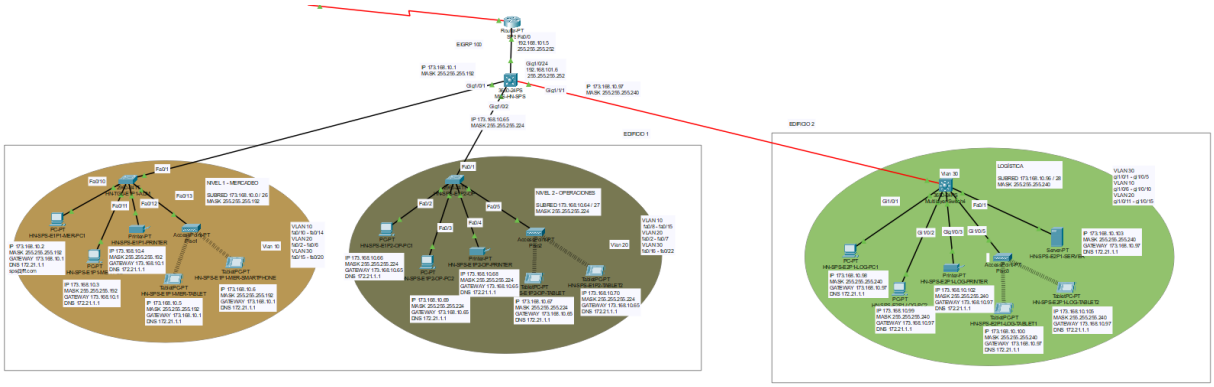


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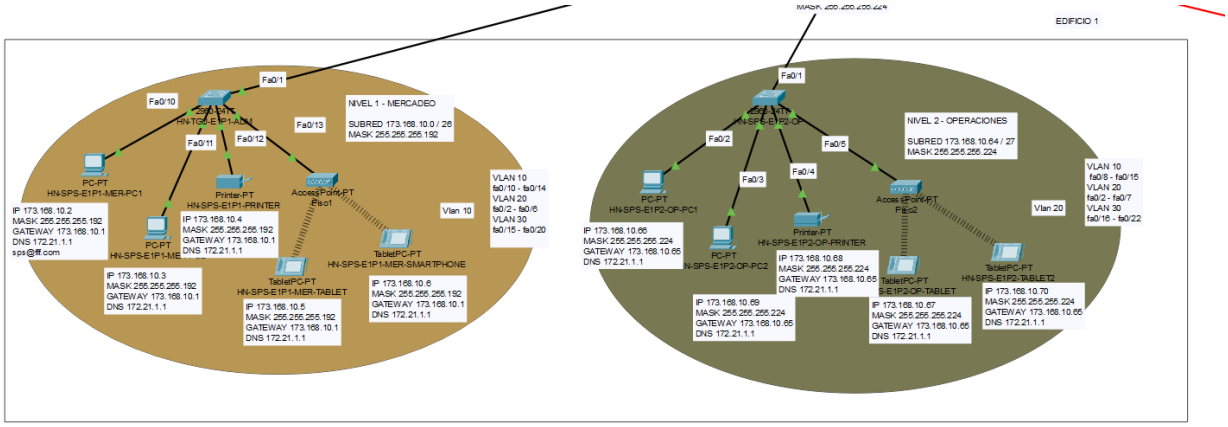


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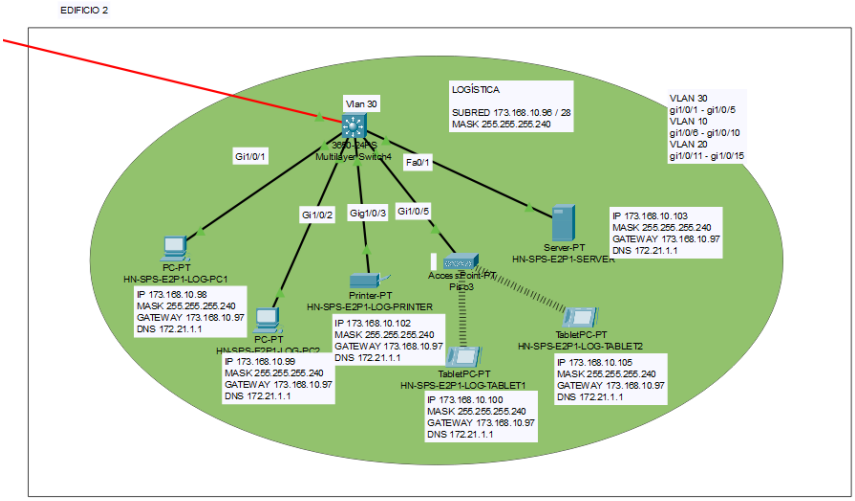
San Pedro Sula (Honduras)



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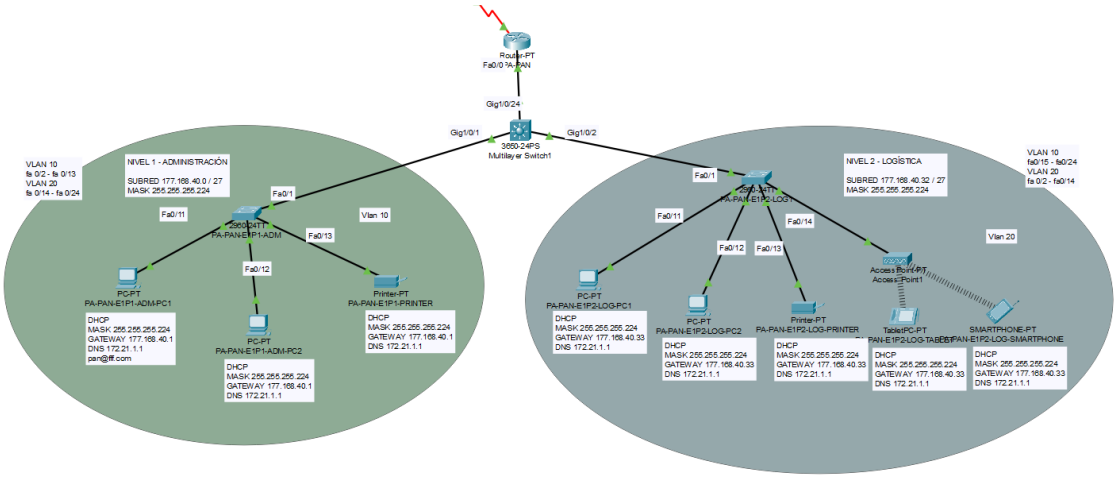


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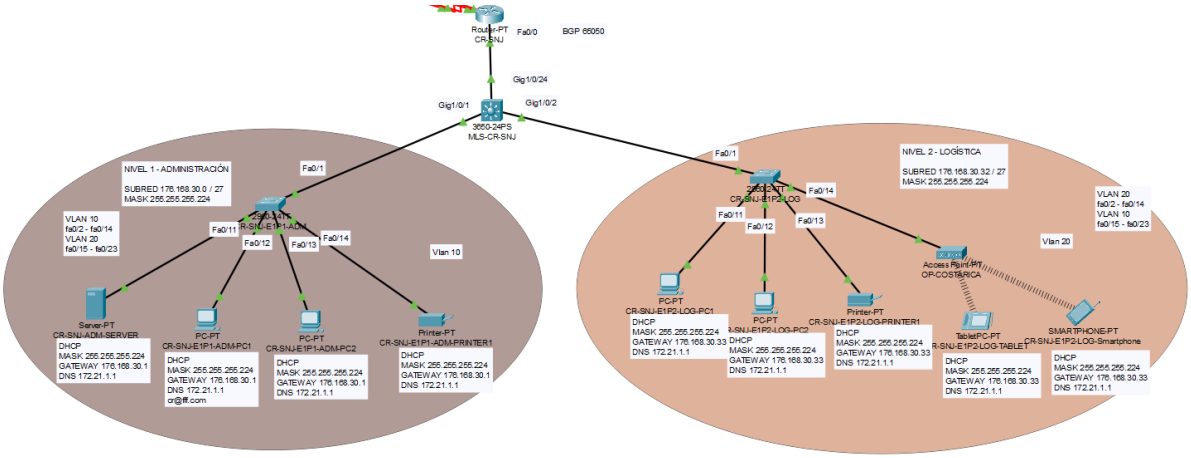
LAN

Ciudad de Panamá (Panamá)



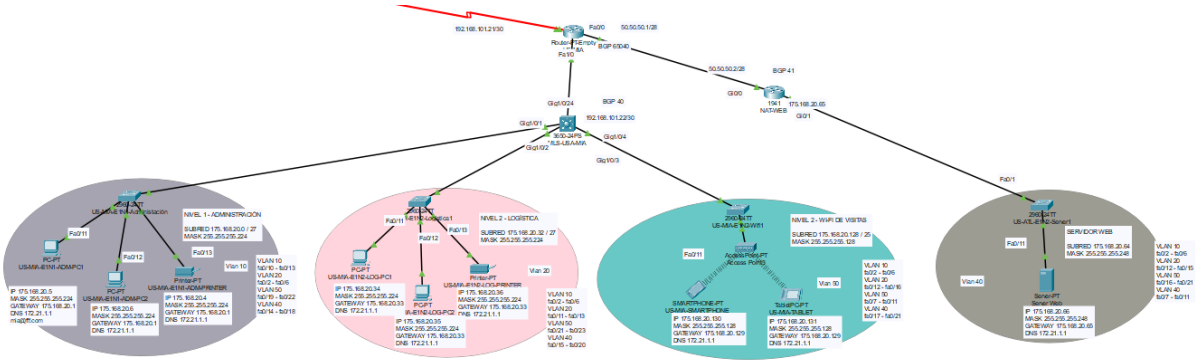
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San José (Costa Rica)



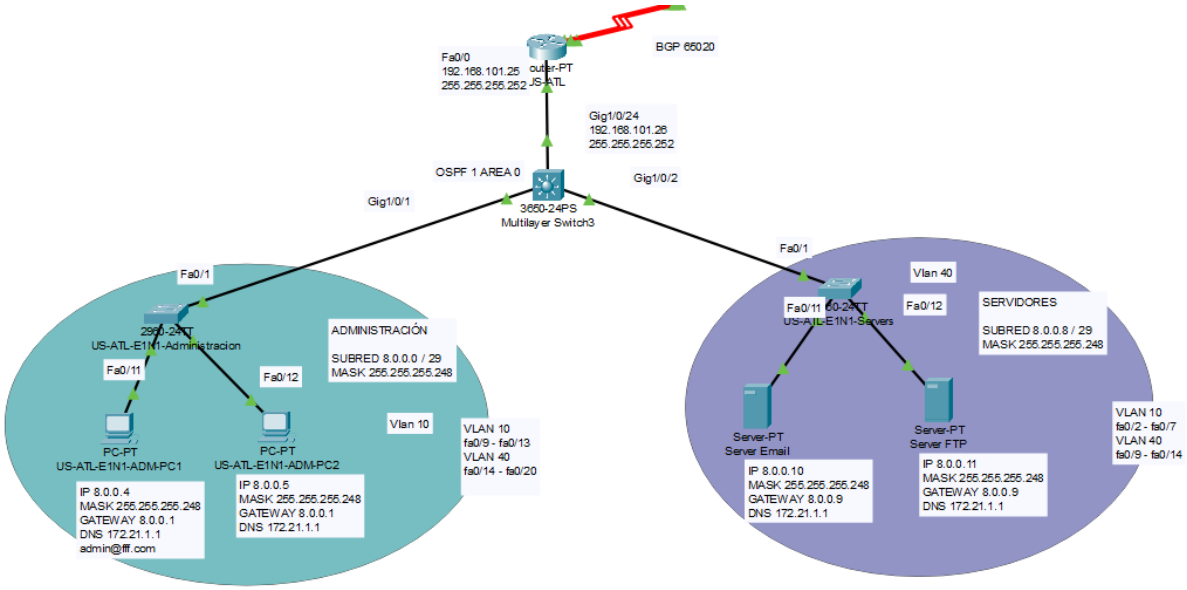
LAN

Miami (Estados Unidos)



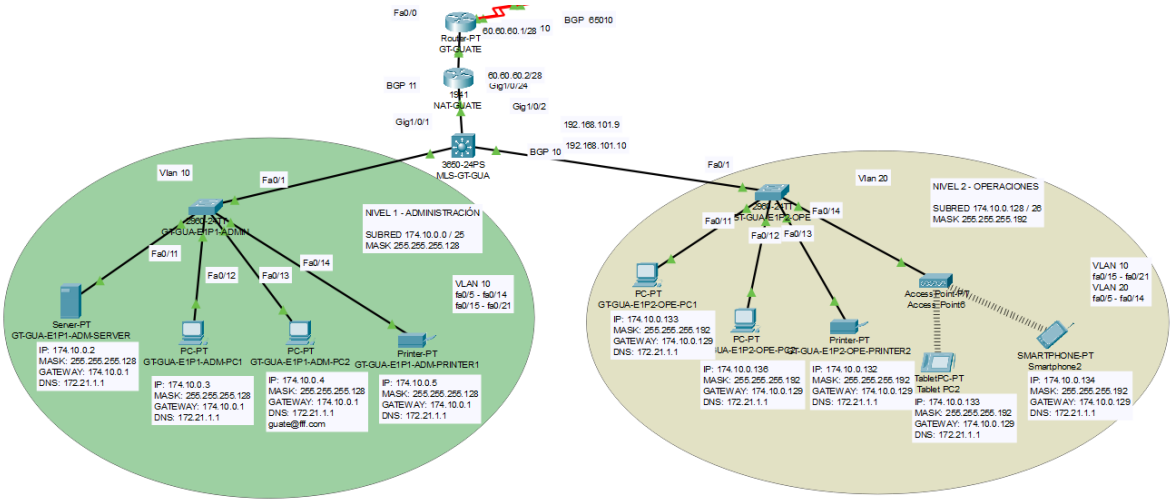
LAN

Atlanta (Estados Unidos)



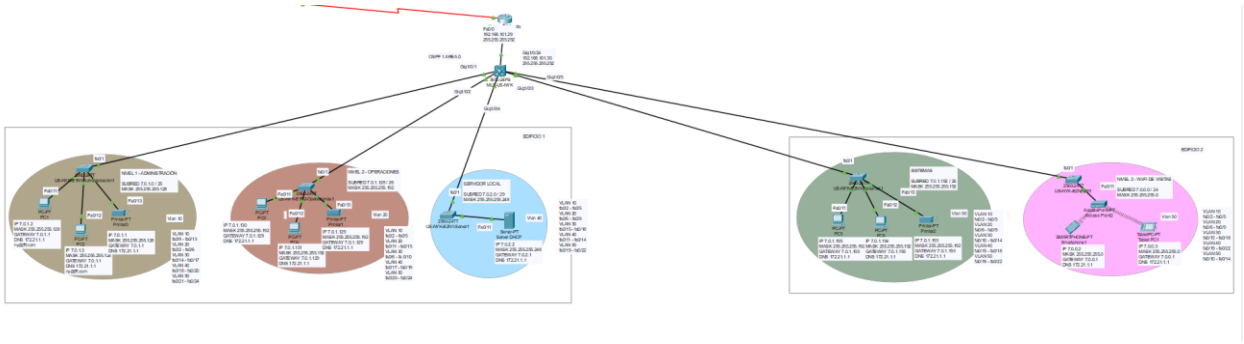
LAN

Guatemala

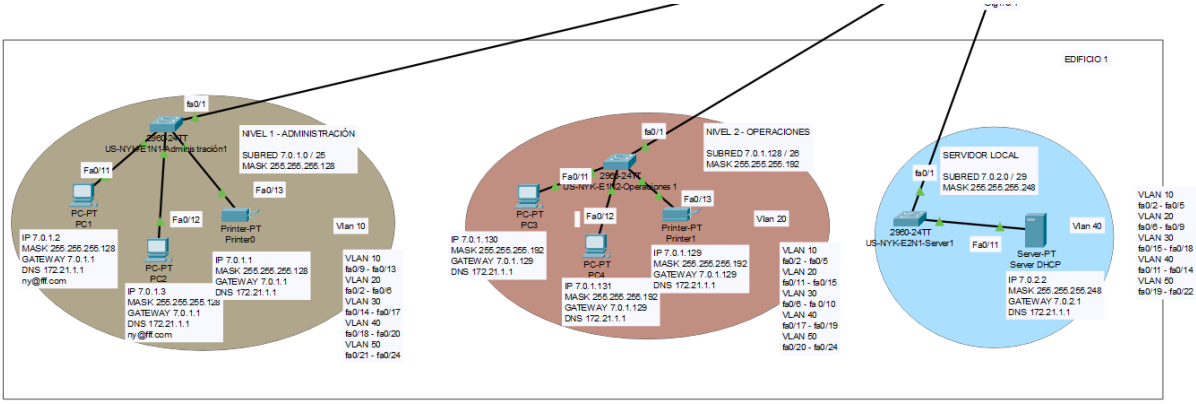


LAN

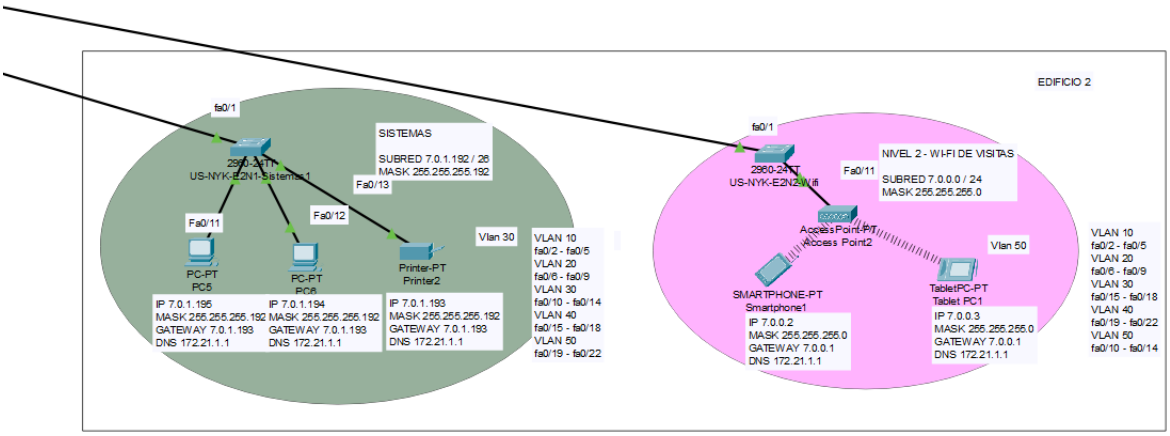
Nueva York (Estados Unidos)



Edificio 1, Nueva York (Estados Unidos)



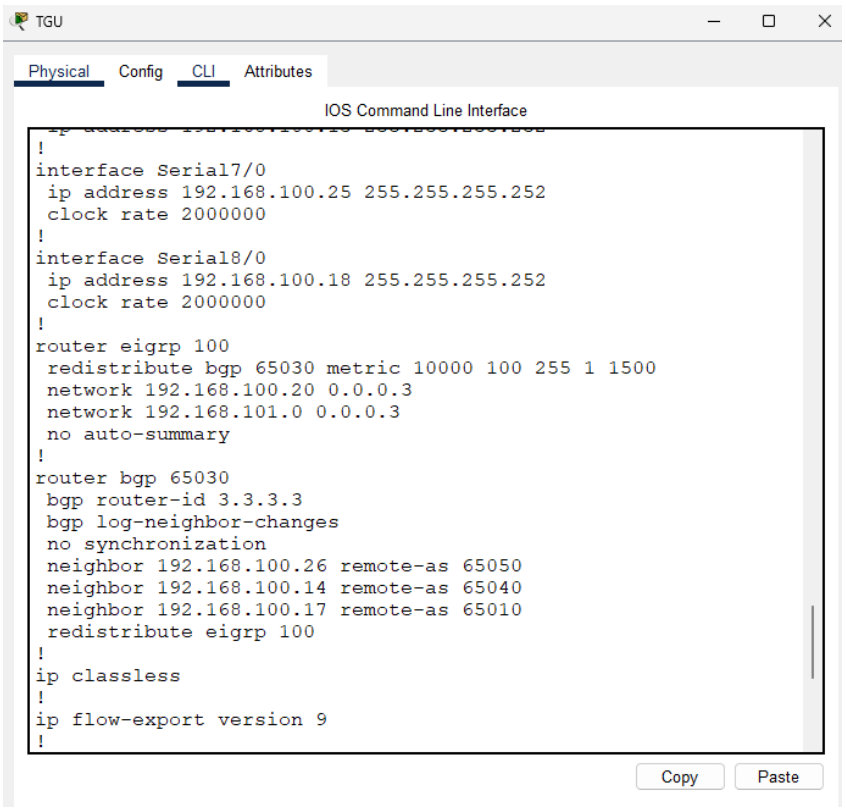
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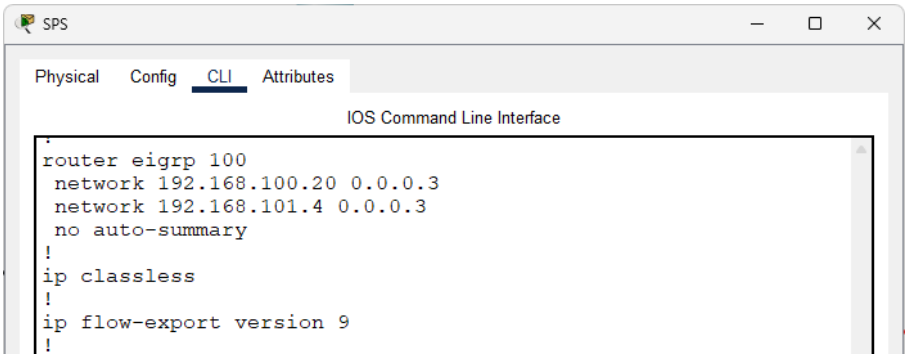
CONFIGURACIÓN DE ROUTERS

PROTOCOLOS

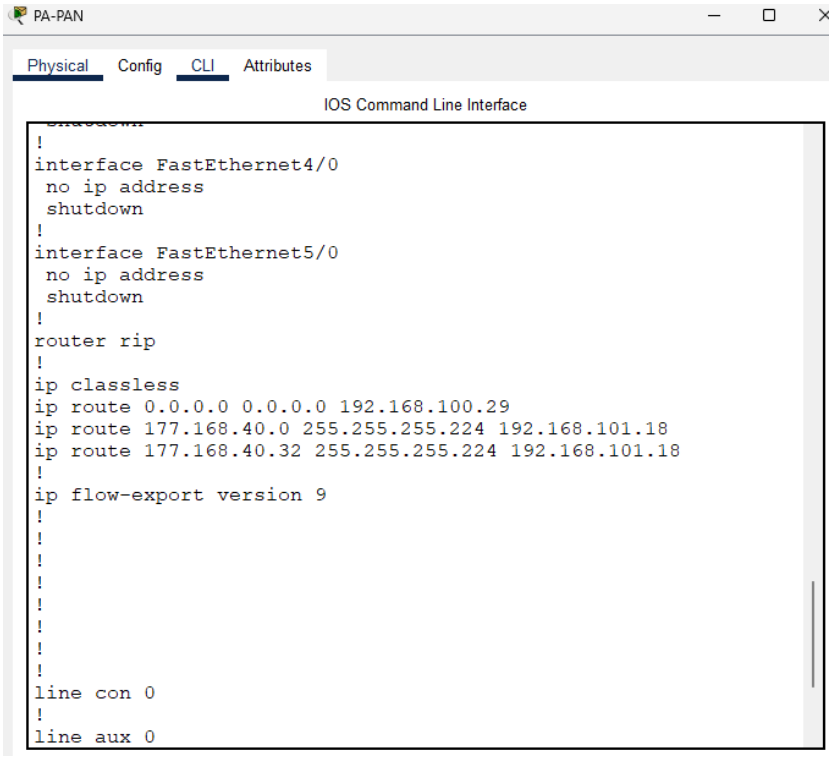
Honduras, Tegucigalpa, EIGRP – BGP



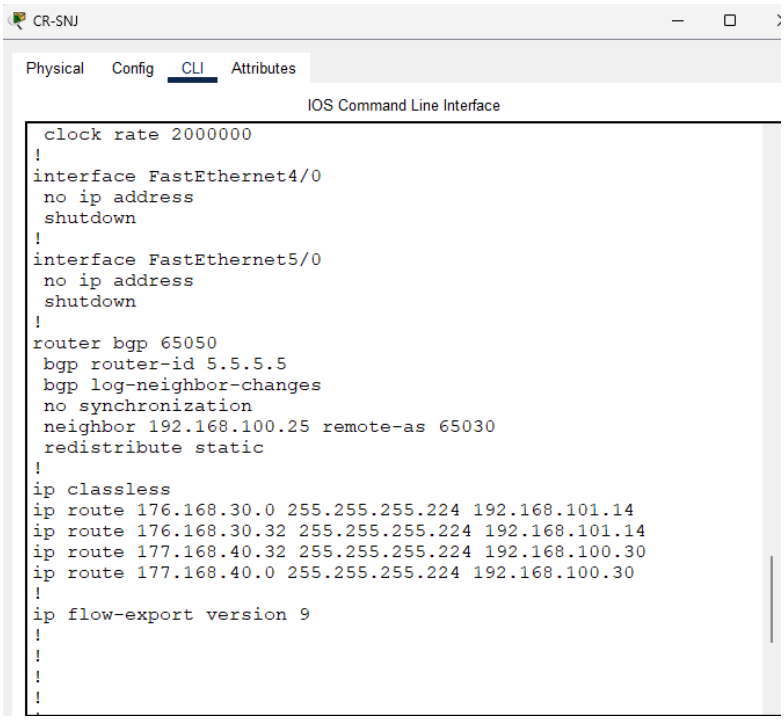
Honduras, San Pedro Sula, EIGRP



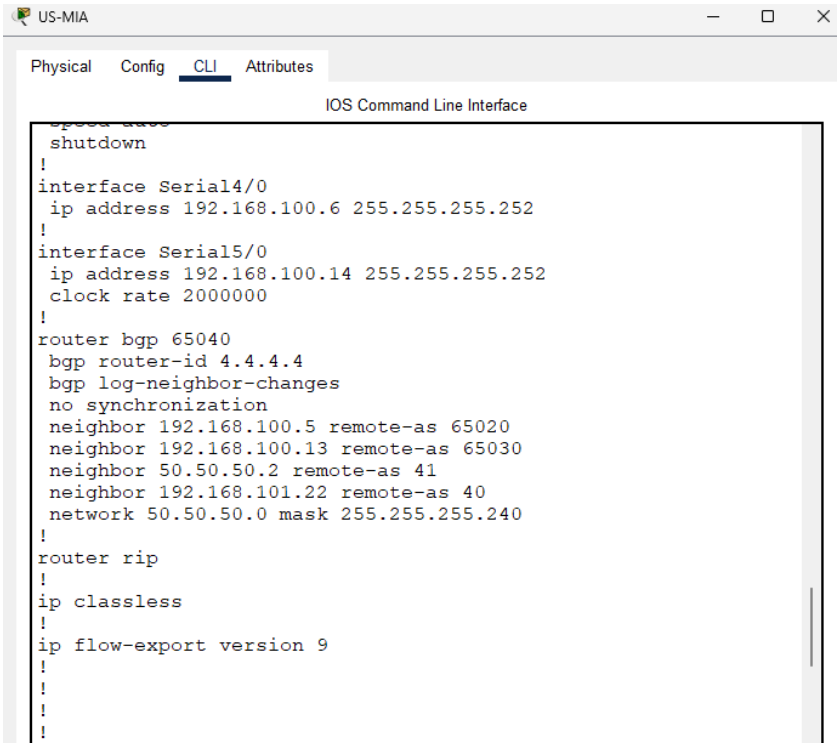
Panamá, Ciudad de Panamá, Estático



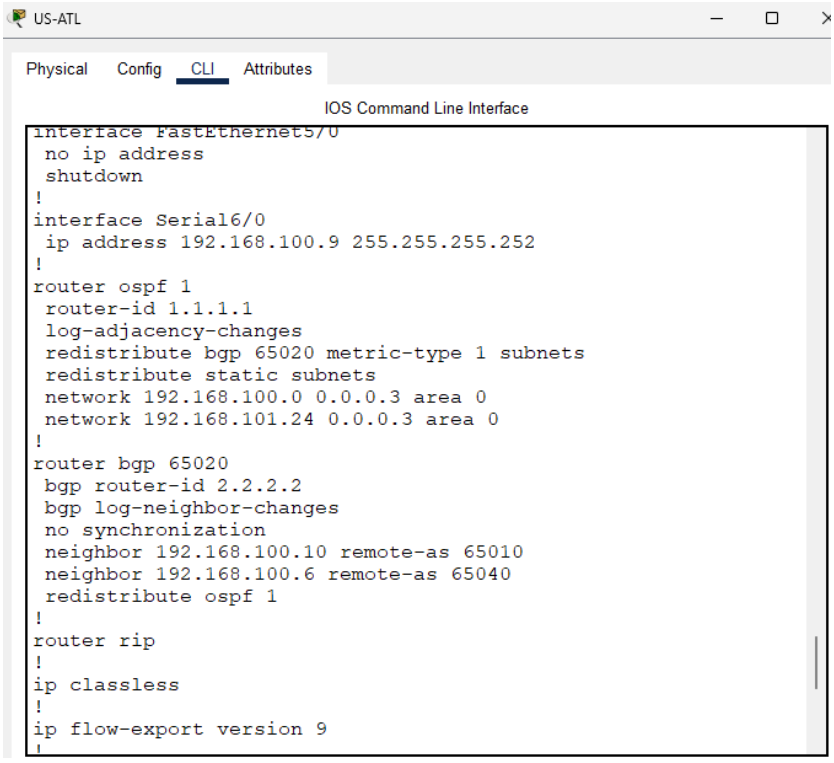
Costa Rica, San José



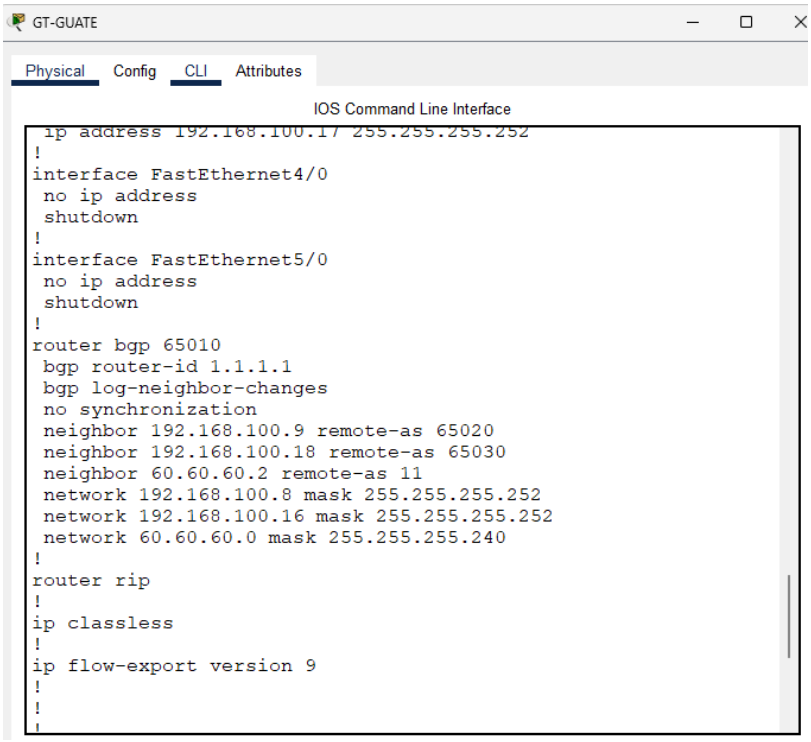
Estados Unidos, Miami



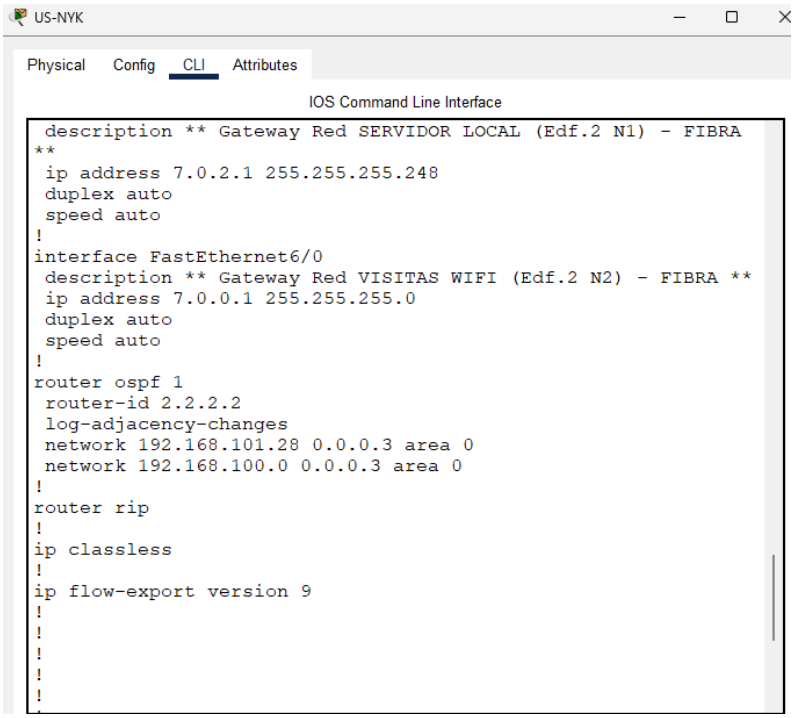
Estados Unidos, Atlanta



Guatemala, BGP



Nueva York, OSPF

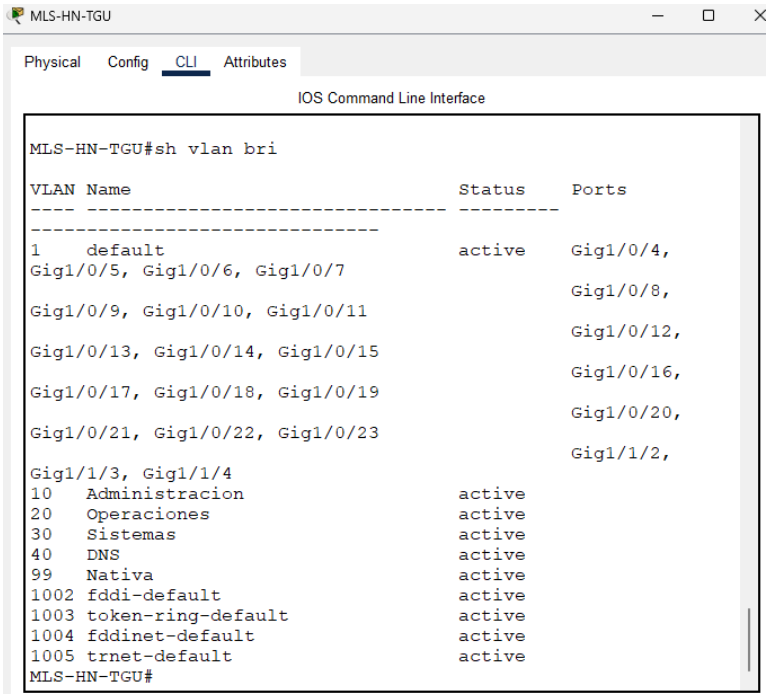


CONFIGURACIÓN DE SWITCHES

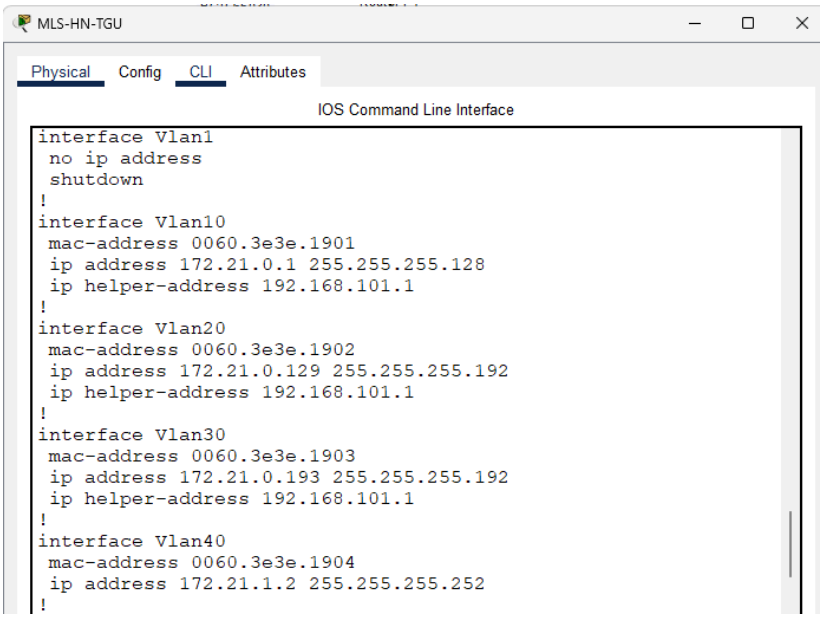
SWITCHES CAPA 3

Tegucigalpa, Honduras

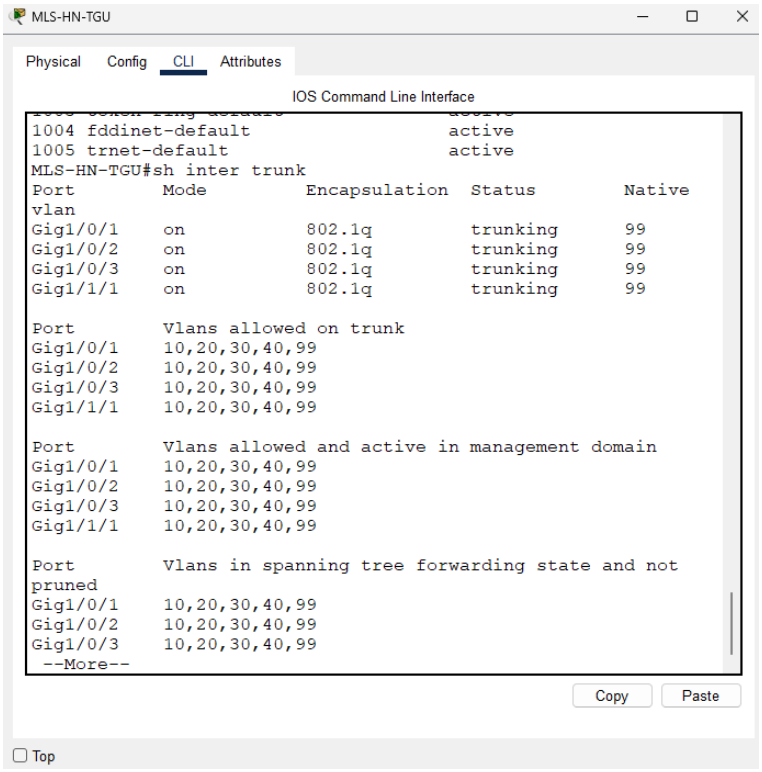
VLANs



SVIs

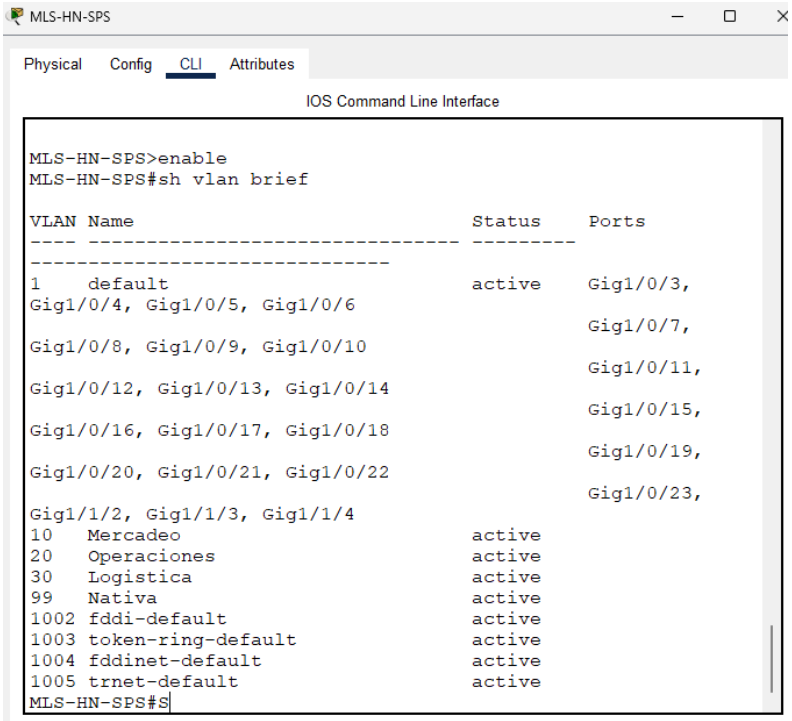


Troncales

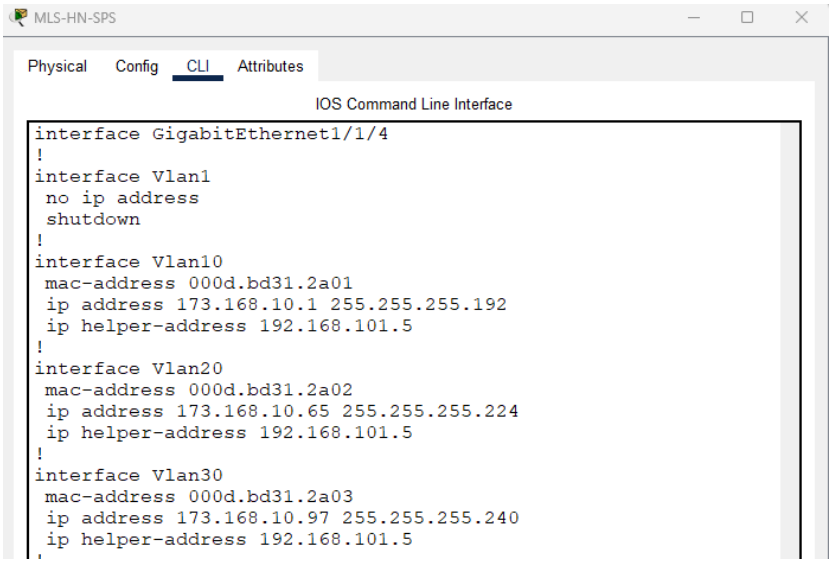


San Pedro Sula, Honduras

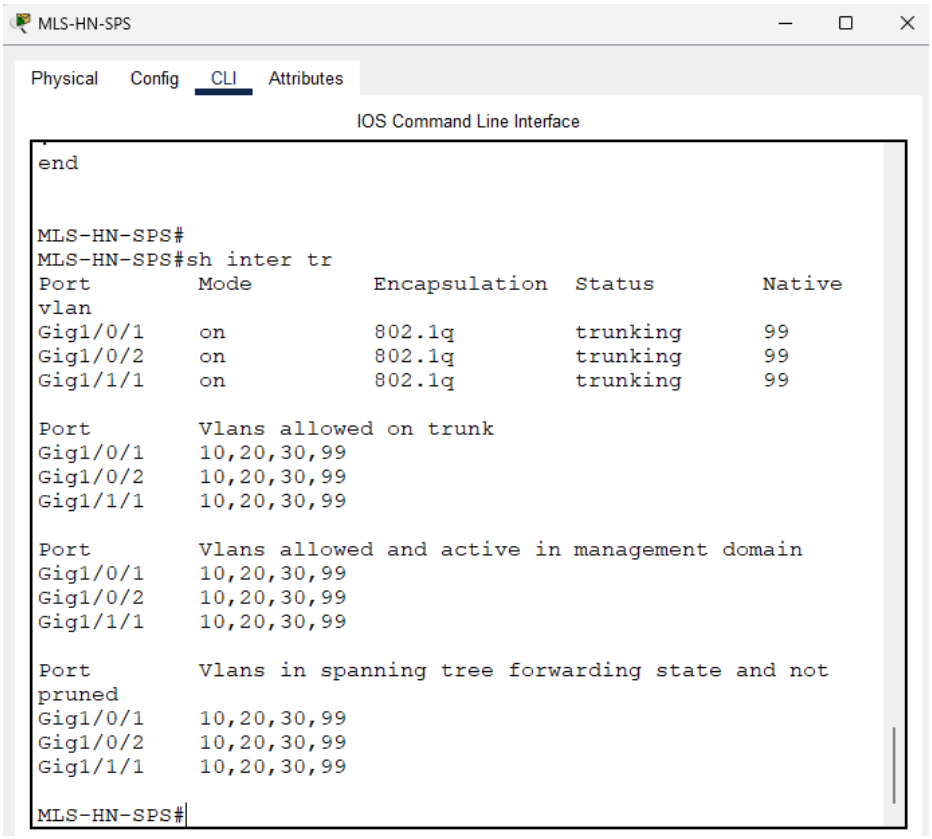
VLANs



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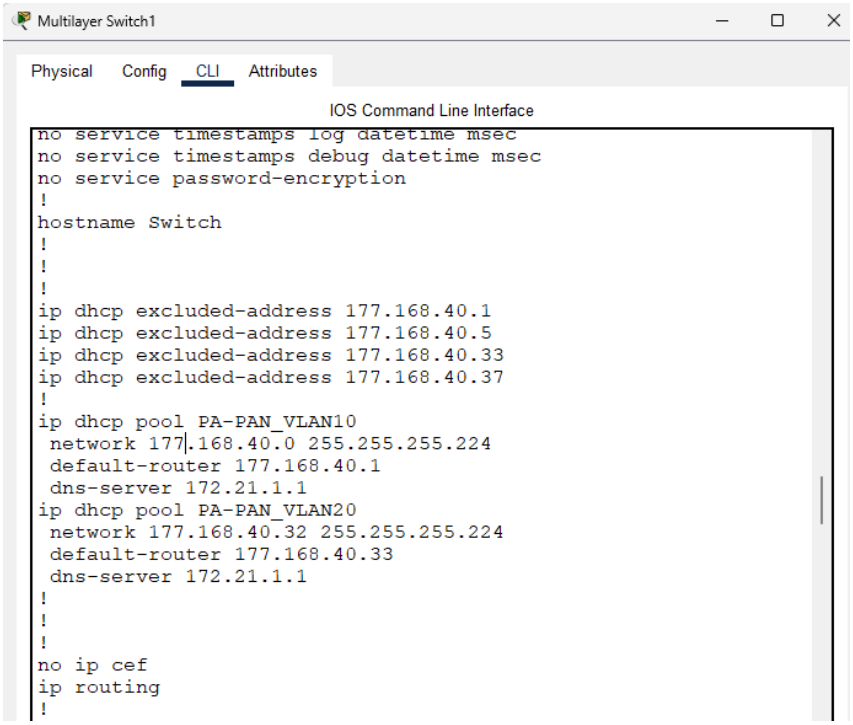


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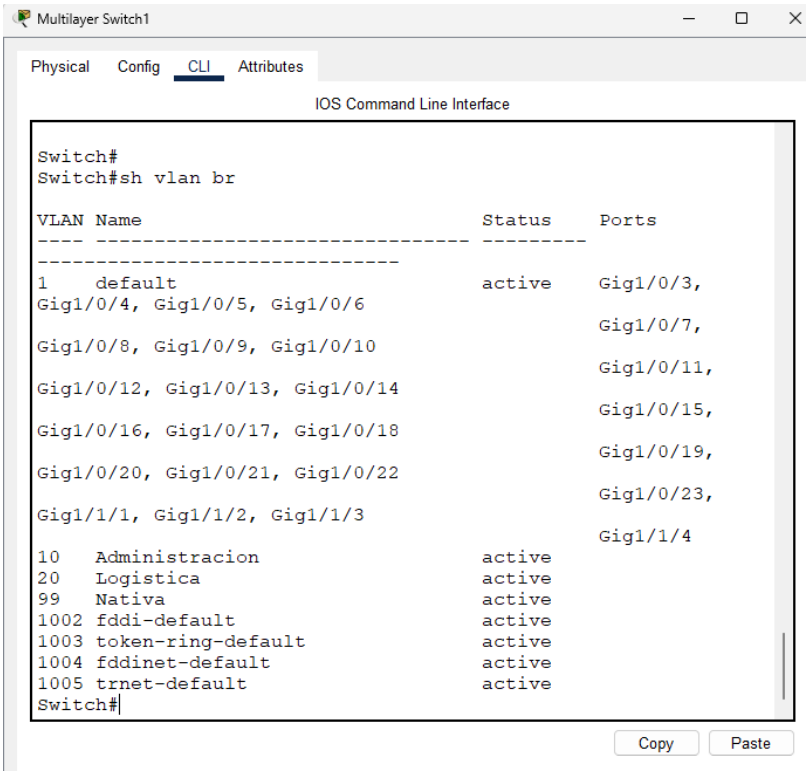


Panamá

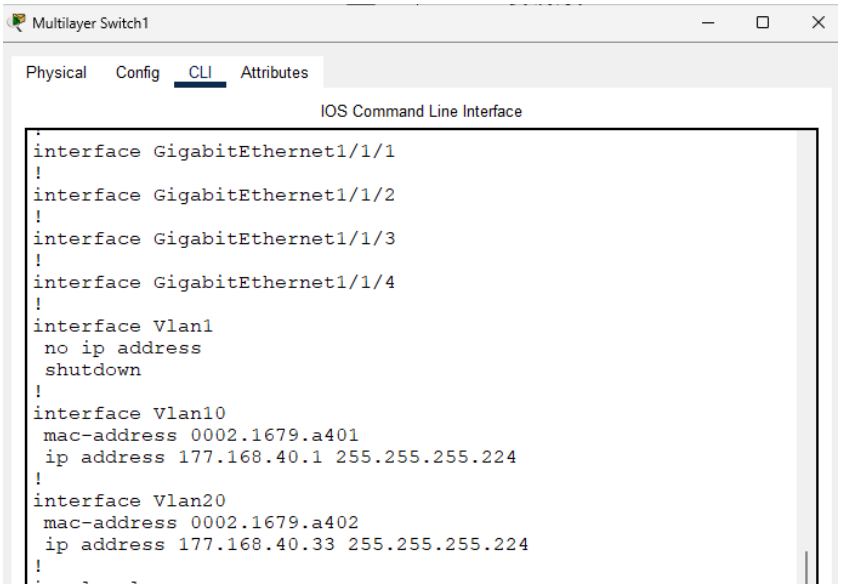
DHCP



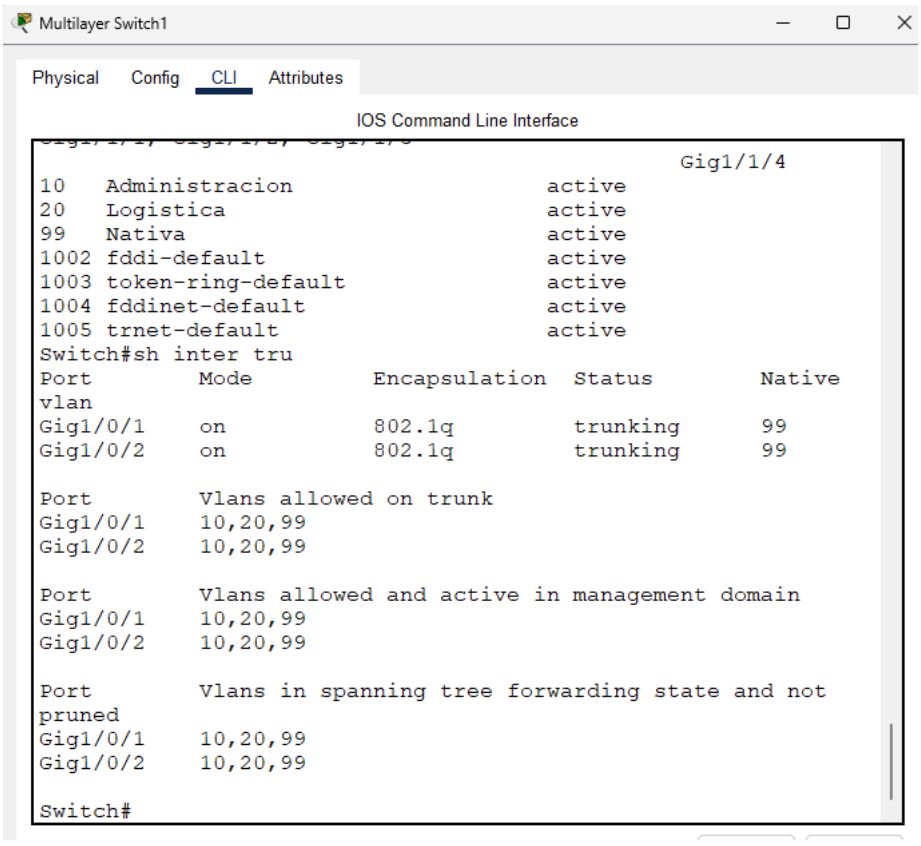
VLANs



SVIs

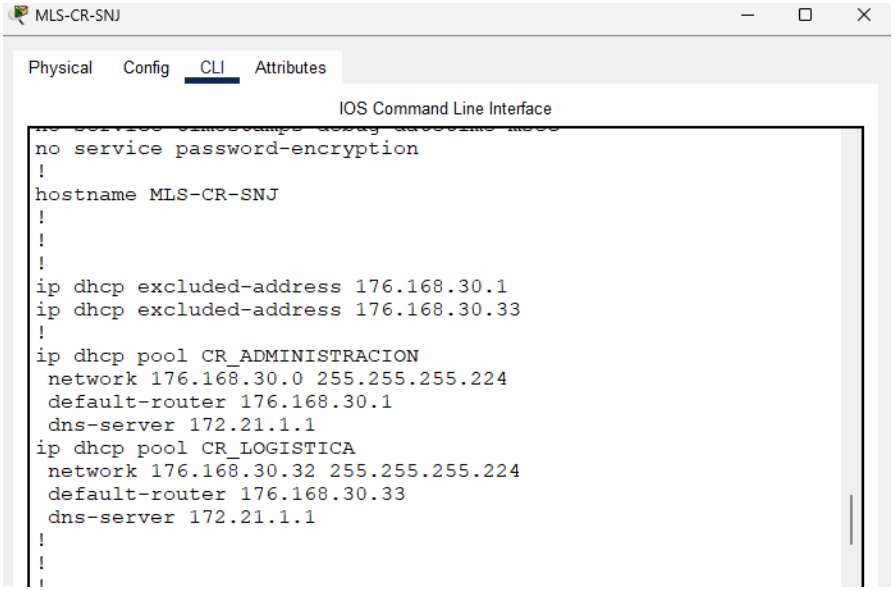


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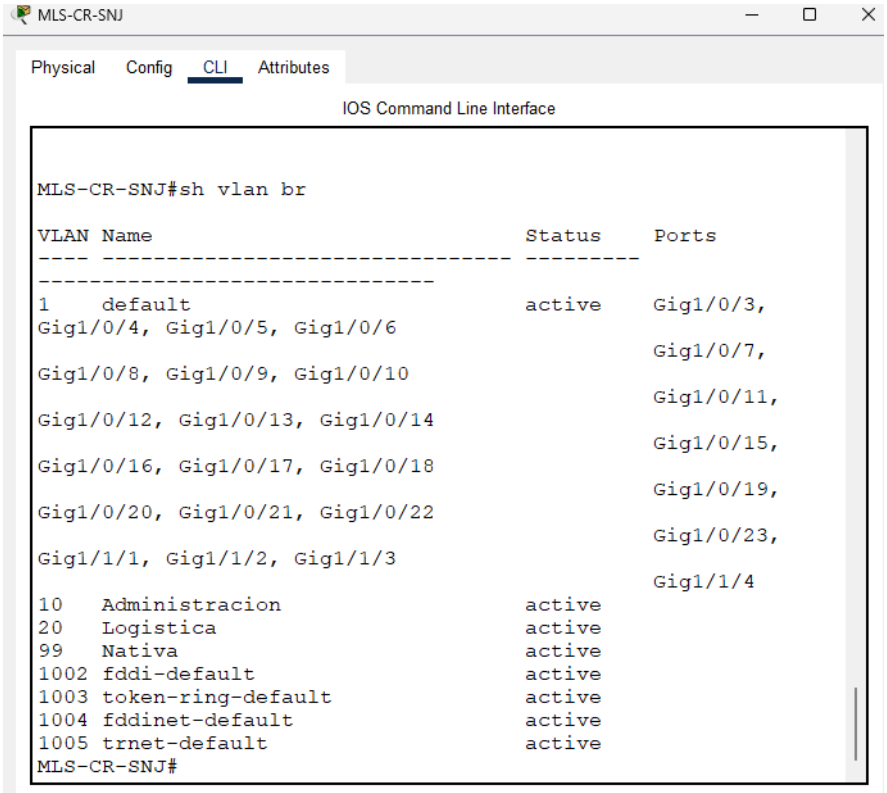


Costa Rica

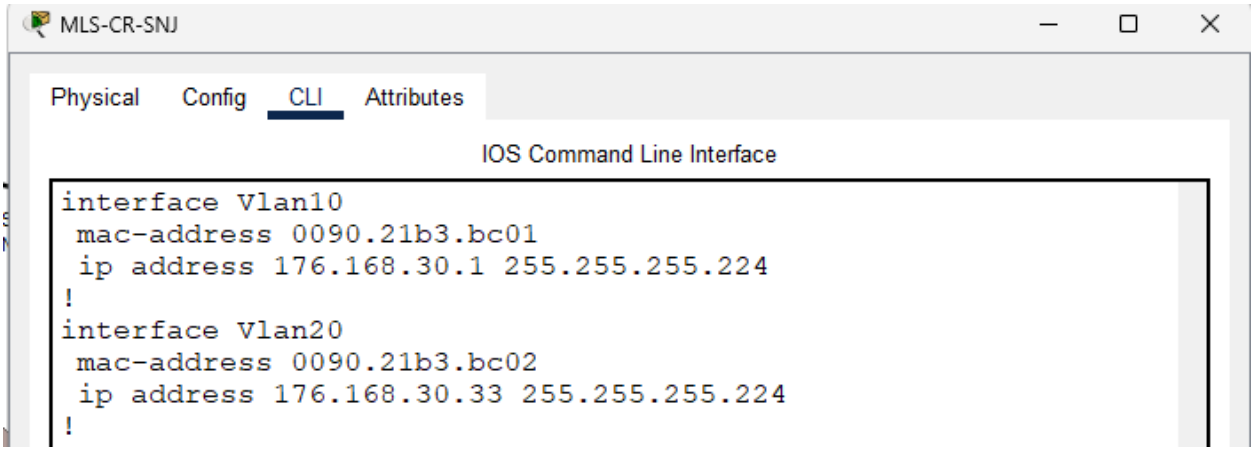
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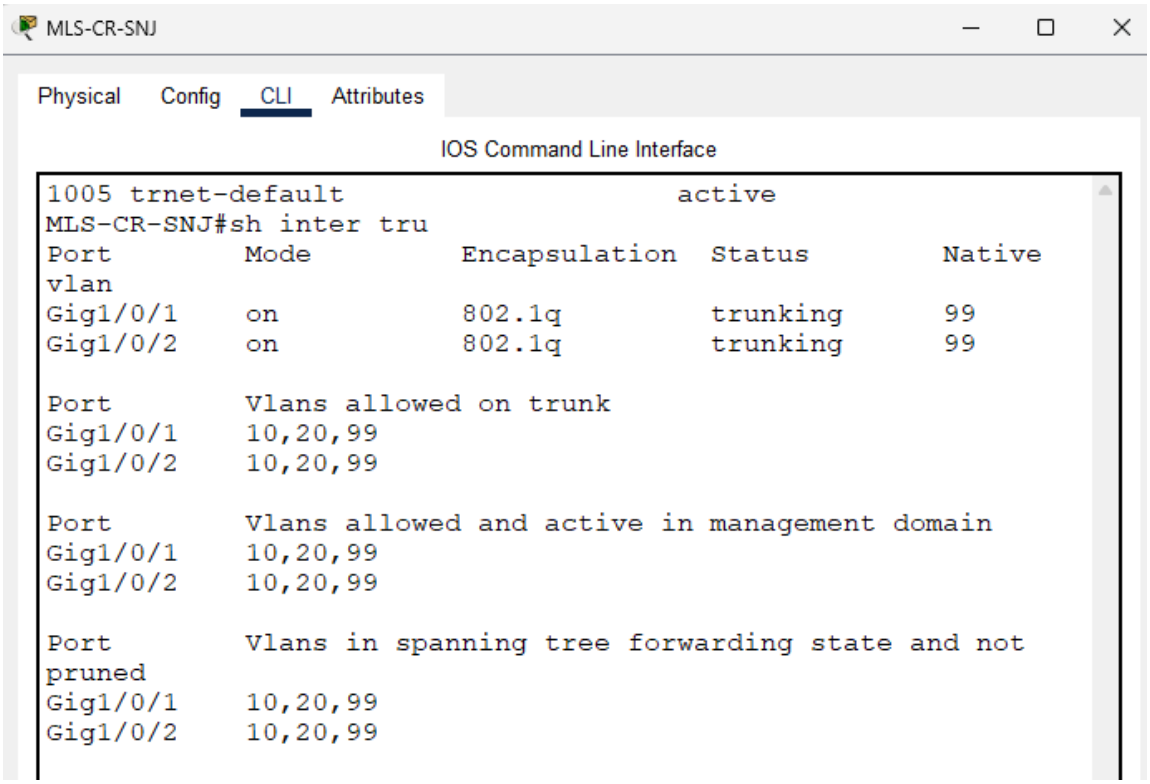
VLANs



SVIs

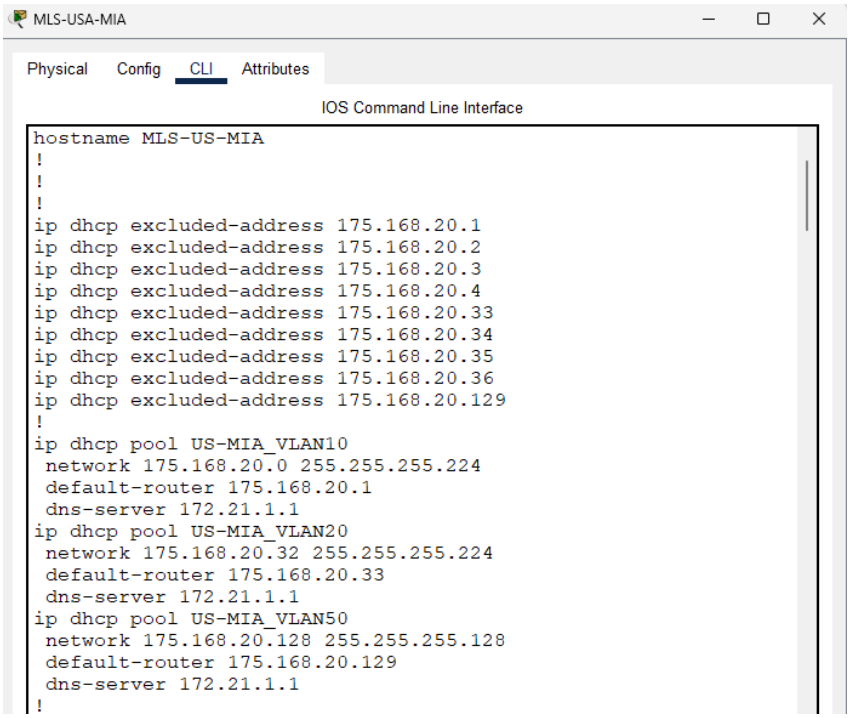


Troncales

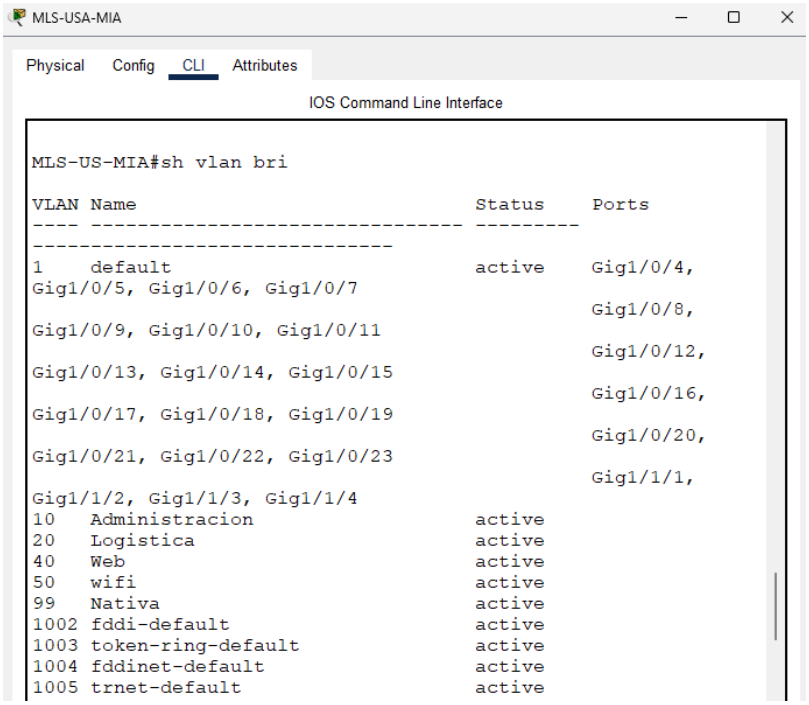


Miami, Estados Unidos

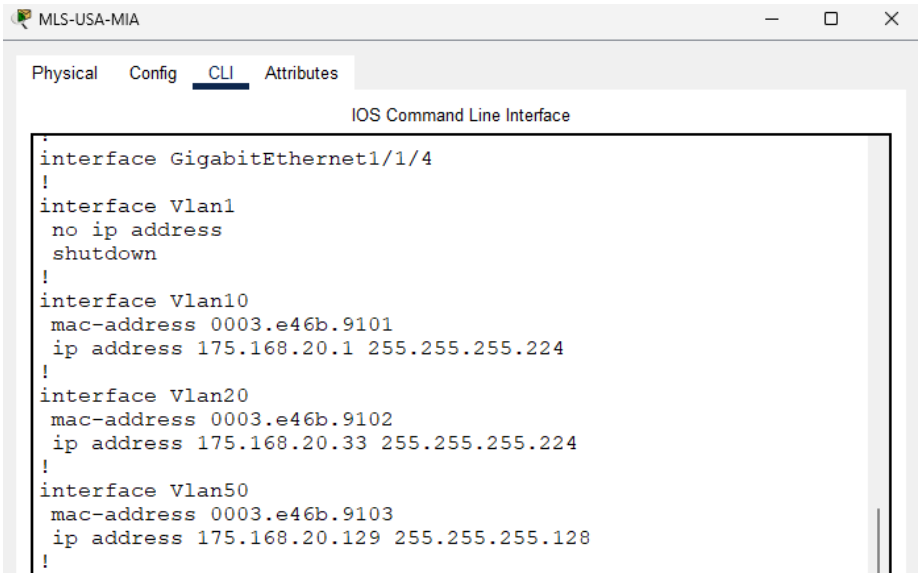
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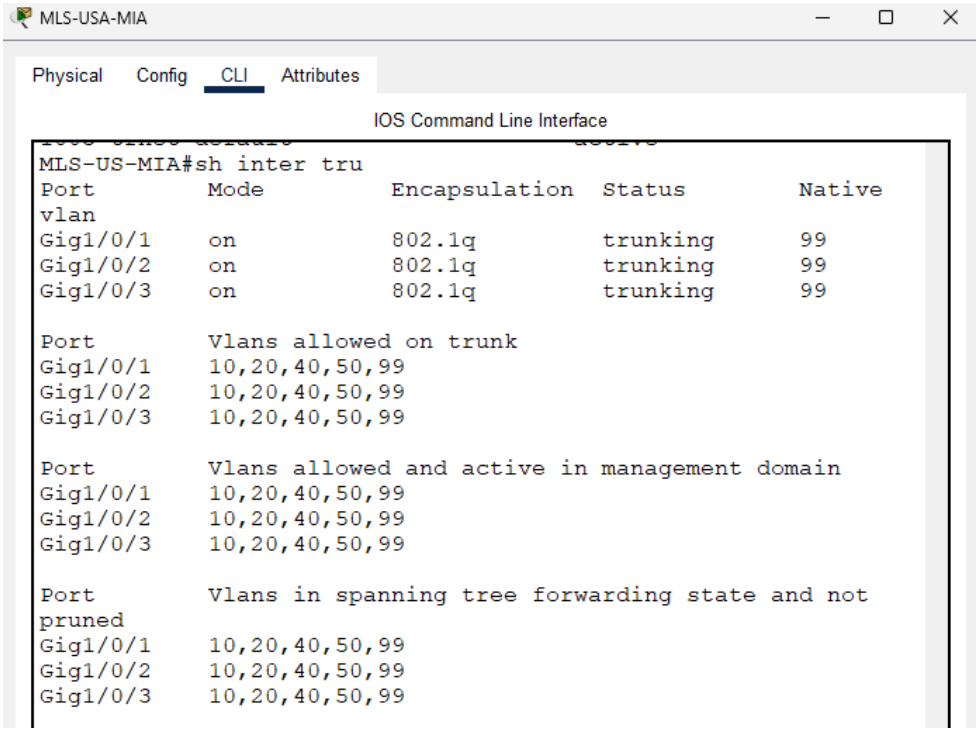
VLANs



SVIs

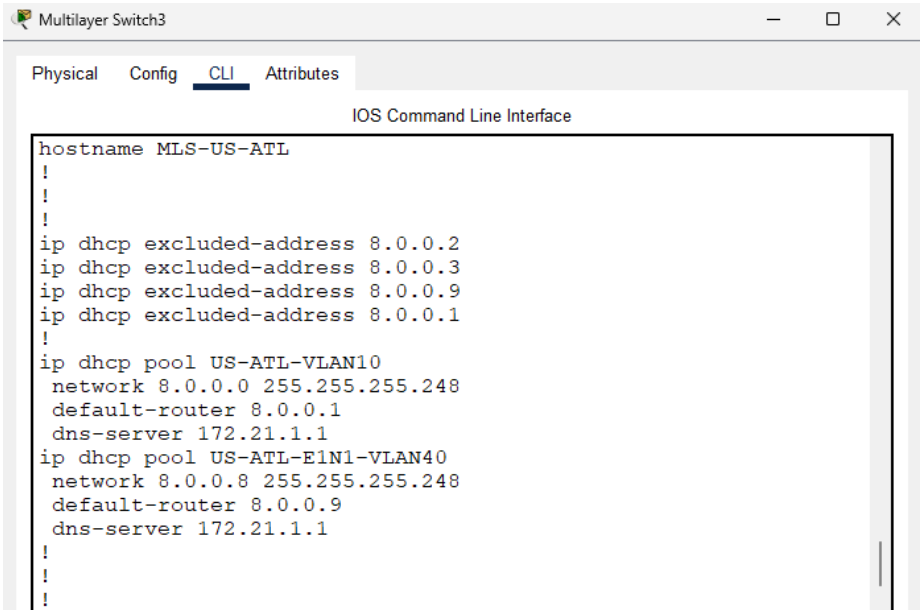


Troncales

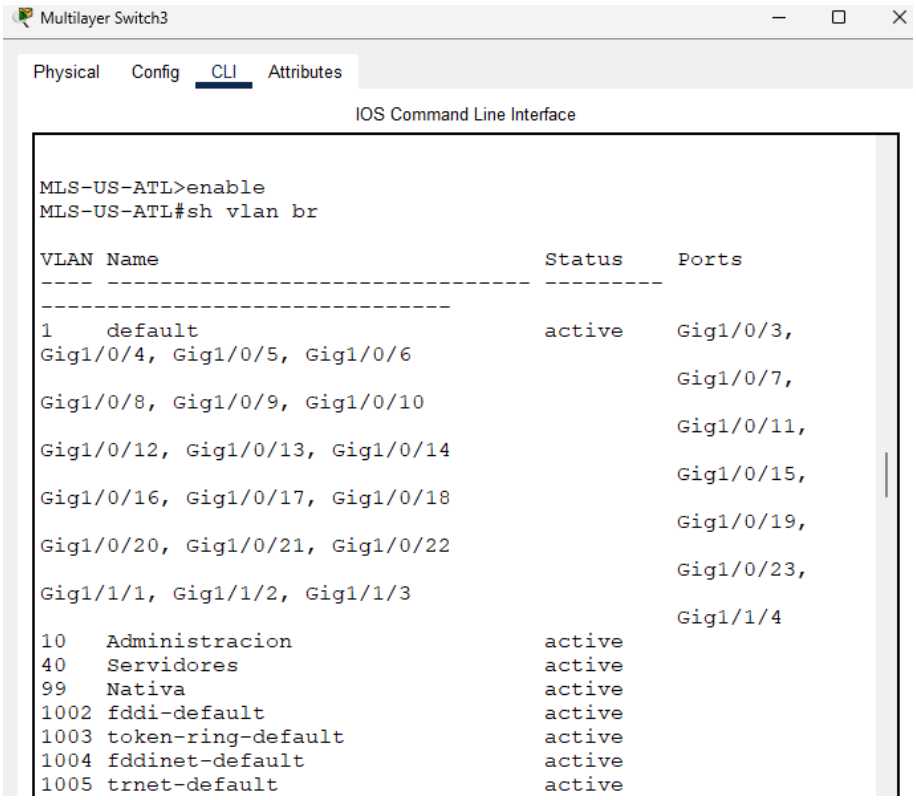


Atlanta, Estados Unidos

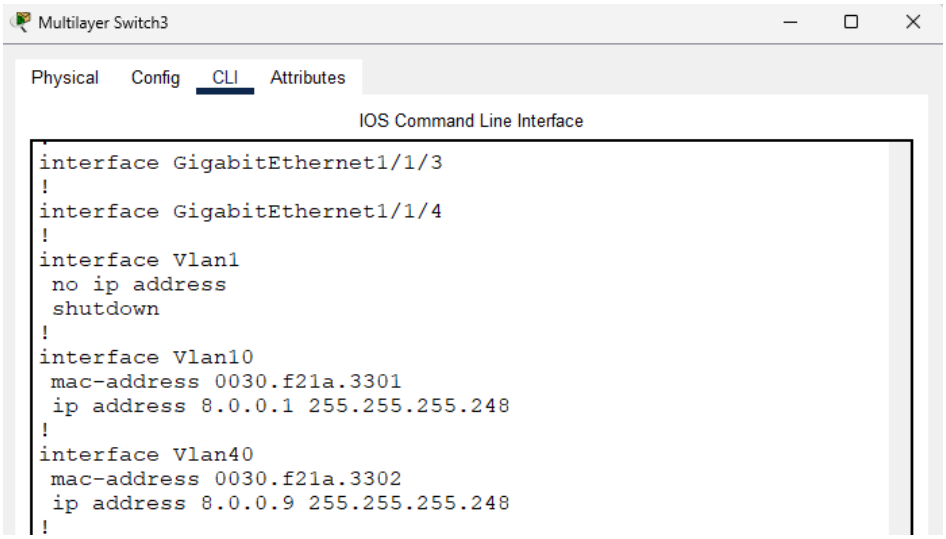
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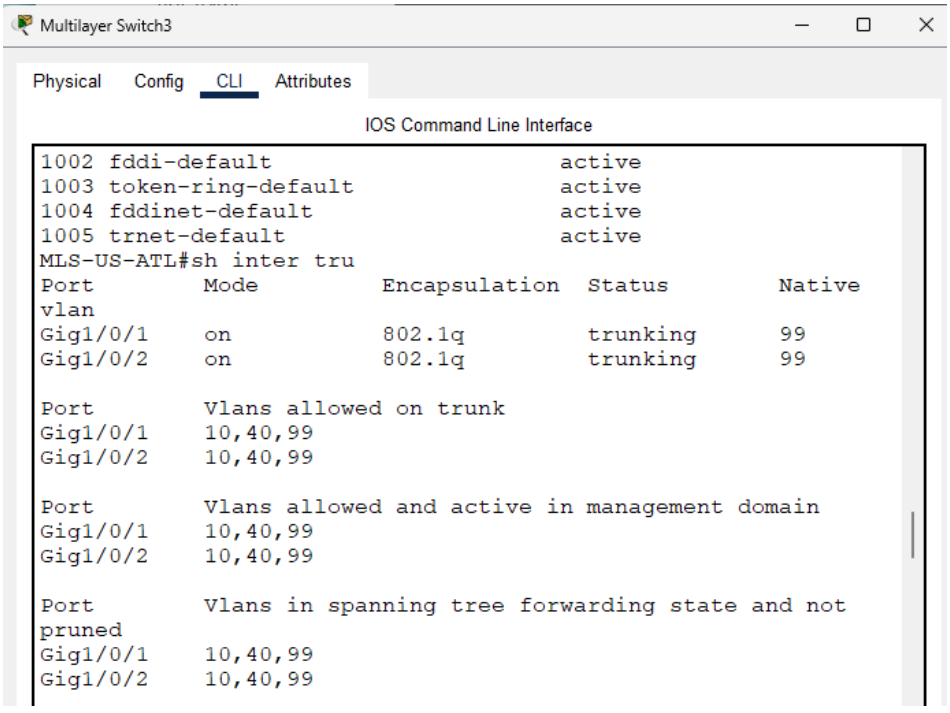
VLANs



SVIs

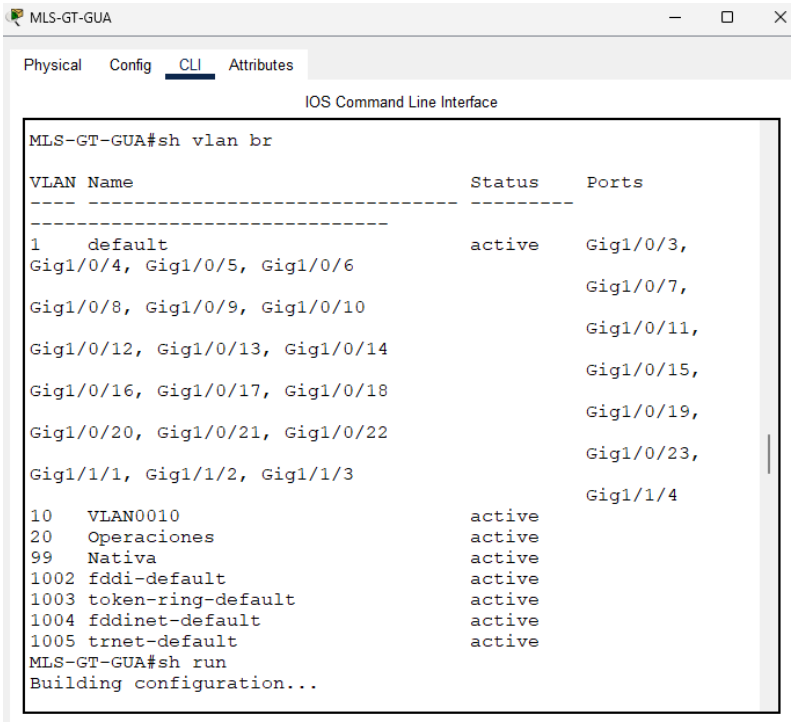


Troncales

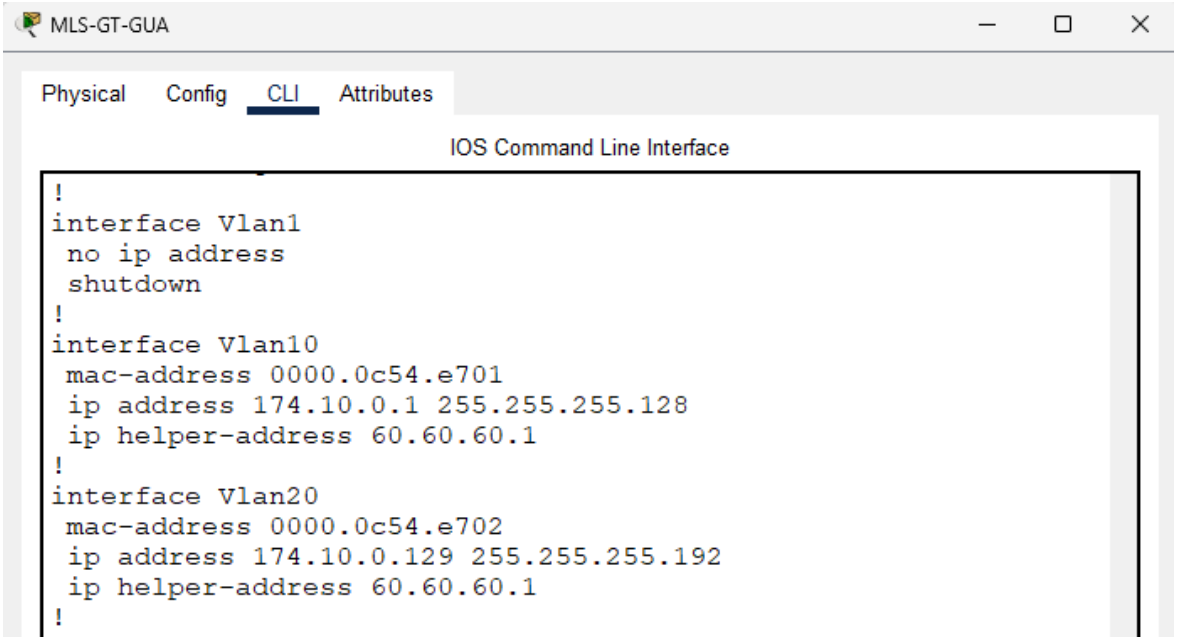


Guatemala

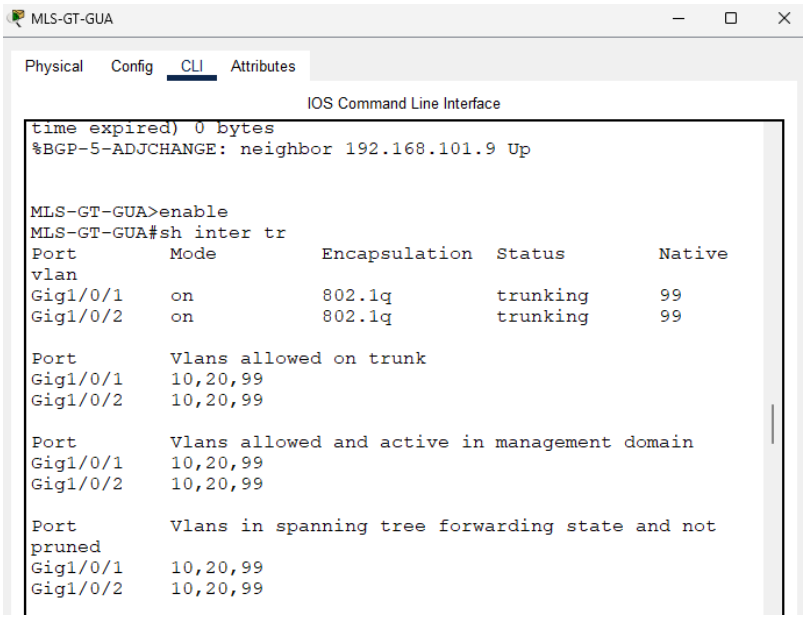
VLANs



SVIs

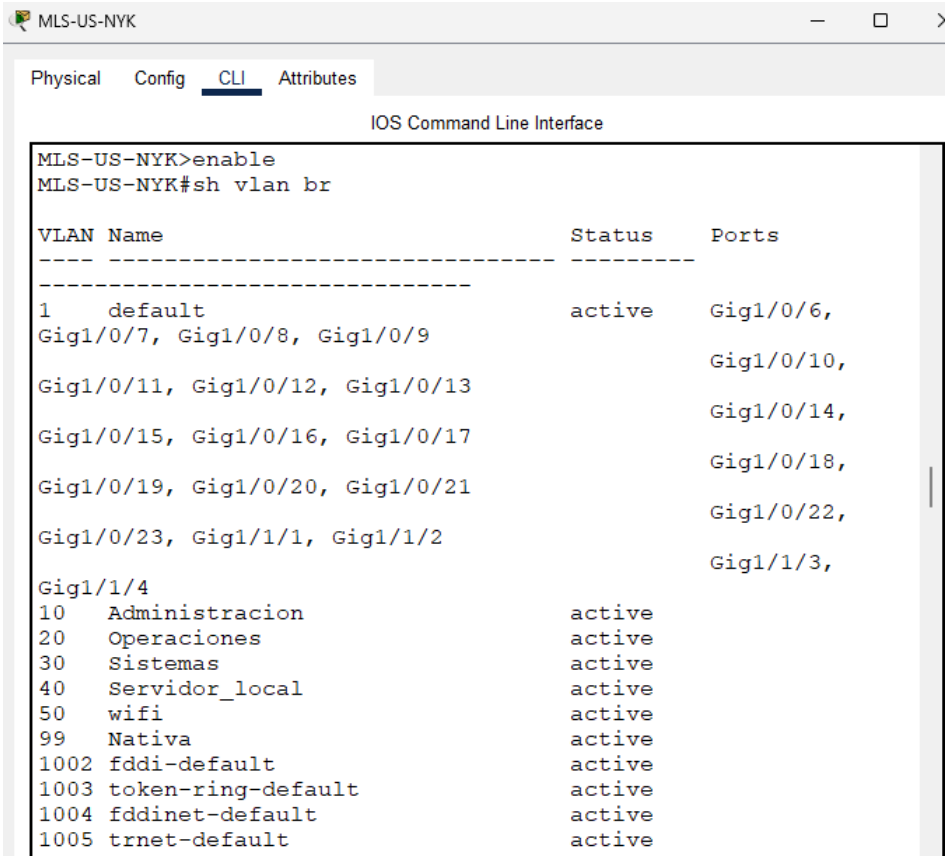


Troncales

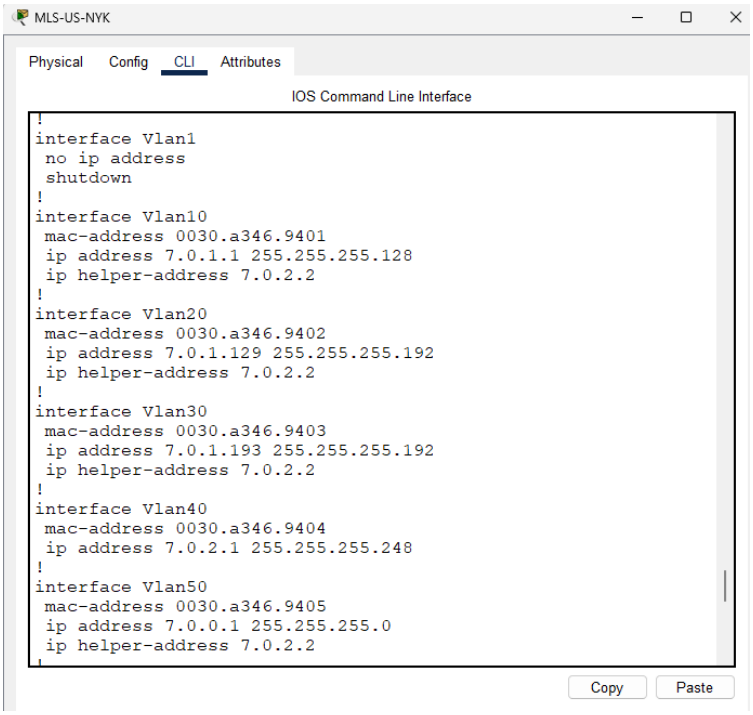


Nueva York

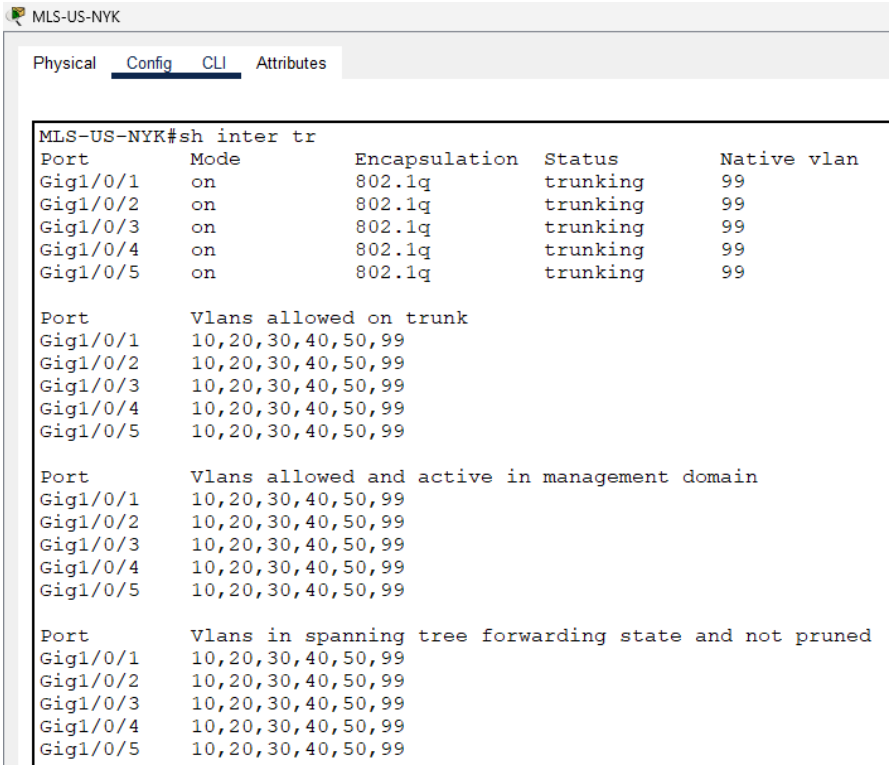
VLANs



SVIs



Troncales



SWITCHES CAPA 2

Tegucigalpa, Honduras

Edificio 1 (1er piso)

```
Switch#sh inter tru
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,99

Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa2/1, Fa3/1, Fa4/1, Fa5/1
10   Administracion                   active
20   Operaciones                      active
30   Sistemas                         active
40   DNS                              active    Fa1/1
99   Nativa                           active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Edificio 1, Administración (1er piso)

```
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40

Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14
20   VLAN0020                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
30   VLAN0030                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19
40   VLAN0040                         active    Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
99   VLAN0099                         active    Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Edificio 1, Operaciones (2do piso)

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
20   VLAN0020                         active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13
30   VLAN0030                         active    Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18
40   VLAN0040                         active    Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
99   VLAN0099                         active    Fa0/7, Fa0/8, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40
```

Edificio 2, Sistemas (1er piso)

```
Switch#sh vlan bri

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/1/2, Gig1/1/3, Gig1/1/4
10   VLAN0010                         active    Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9
                                                Gig1/0/10, Gig1/0/11
20   VLAN0020                         active    Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15
30   VLAN0030                         active    Gig1/0/1, Gig1/0/2, Gig1/0/3, Gig1/0/4
                                                Gig1/0/5
40   VLAN0040                         active    Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19
                                                Gig1/0/20
99   VLAN0099                         active    Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Gig1/1/1    on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gig1/1/1    10,20,30,40,50

Port        Vlans allowed and active in management domain
Gig1/1/1    10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gig1/1/1    10,20,30,40
```

San Pedro Sula (Honduras)

Edificio 1, Mercadeo (1er piso)

```
Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14
20   VLAN0020                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
30   VLAN0030                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
40   VLAN0040                         active
99   VLAN0099                         active    Fa0/7, Fa0/8, Fa0/9, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,99

Switch#
```

Edificio 1, Operaciones (2do piso)

```
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,99

Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
20   VLAN0020                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7
30   VLAN0030                         active    Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22
99   VLAN0099                         active    Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Edificio 2, Logística (1er piso)

```
Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/1/2, Gig1/1/3, Gig1/1/4
10   VLAN0010                         active    Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9
                                                Gig1/0/10
20   VLAN0020                         active    Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14
                                                Gig1/0/15
30   VLAN0030                         active    Gig1/0/1, Gig1/0/2, Gig1/0/3, Gig1/0/4
                                                Gig1/0/5
99   VLAN0099                         active    Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19
                                                Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23
                                                Gig1/0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Gig1/1/1    on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gig1/1/1    10,20,30,40,50

Port        Vlans allowed and active in management domain
Gig1/1/1    10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig1/1/1    10,20,30
```

Ciudad de Panamá (Panamá)

Edificio 1, Administración (1er piso)

```
Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
20   VLAN0020                         active    Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
30   VLAN0030                         active
99   VLAN0099                         active    Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30
```

Edificio 1, Logística (2do piso)

```
Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
20   VLAN0020                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14
99   VLAN0099                         active    Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20
```

San José (Costa Rica)

Edificio 1, Administración (1er piso)

```
Switch>enable
Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14
20   VLAN0020                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23
99   VLAN0099                         active    Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20
```

Edificio 1, Logística (2do piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
20   VLAN0020                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14
99   VLAN0099                         active    Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20
```

Miami (Estados Unidos)

Edificio 1, Administración (1er piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
20   VLAN0020                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
30   VLAN0030                         active
40   VLAN0040                         active    Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18
50   VLAN0050                         active    Fa0/19, Fa0/20, Fa0/21, Fa0/22
99   VLAN0099                         active    Fa0/7, Fa0/8, Fa0/9, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Edificio 1, Logística (2do piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
20   VLAN0020                         active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14
30   VLAN0030                         active
40   VLAN0040                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
50   VLAN0050                         active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
99   VLAN0099                         active    Fa0/7, Fa0/8, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Edificio 1, Visitas (2do piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
20   VLAN0020                         active    Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16
30   VLAN0030                         active
40   VLAN0040                         active    Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21
50   VLAN0050                         active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11
99   VLAN0099                         active    Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inte tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Edificio 1, Servidor Web (2do piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
20   VLAN0020                         active    Fa0/12, Fa0/13, Fa0/14, Fa0/15
30   VLAN0030                         active
40   VLAN0040                         active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11
50   VLAN0050                         active    Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21
99   VLAN0099                         active    Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Atlanta (Estados Unidos)

Edificio 1, Administración (1er piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13
20   VLAN0020                         active
30   VLAN0030                         active
40   VLAN0040                         active    Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20
99   VLAN0099                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,99
```

Edificio 1, Servidores (1er piso)

```
Switch>sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active
10   VLAN0010                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                 Fa0/6, Fa0/7
20   VLAN0020                          active
30   VLAN0030                          active
40   VLAN0040                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14
99   VLAN0099                          active    Fa0/8, Fa0/15, Fa0/16, Fa0/17
                                                 Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                 Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                 Gig0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>sh inte tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       40

Port        Vlans allowed and active in management domain
Fa0/1       40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       40
```

Guatemala

Edificio 1, Administración (1er piso)

```
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,99

Switch>sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active
10   VLAN0010                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14
20   VLAN0020                          active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21
30   VLAN0030                          active
40   VLAN0040                          active
99   VLAN0099                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/22
                                                 Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>
```

Edificio 1, Operaciones (2do piso)

```
Switch>sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active
10   VLAN0010                          active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21
20   VLAN0020                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14
30   VLAN0030                          active
40   VLAN0040                          active
99   VLAN0099                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/22
                                                 Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,99
```

Nueva York

Edificio 1, Administración (1er piso)

```
Switch>sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active
10   VLAN0010                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13
20   VLAN0020                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                 Fa0/6
30   VLAN0030                          active    Fa0/14, Fa0/15, Fa0/16, Fa0/17
40   VLAN0040                          active    Fa0/18, Fa0/19, Fa0/20
50   VLAN0050                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
99   VLAN0099                          active    Fa0/7, Fa0/8, Gig0/1, Gig0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Edificio 1, Operaciones (2do piso)

```
Switch>sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active
10   VLAN0010                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
20   VLAN0020                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                 Fa0/15
30   VLAN0030                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                 Fa0/10
40   VLAN0040                          active    Fa0/16, Fa0/17, Fa0/18, Fa0/19
50   VLAN0050                          active    Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                 Fa0/24
99   VLAN0099                          active    Gig0/1, Gig0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Edificio 1, Servidor Local (2do piso)

```
Switch>sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active
10   VLAN0010                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
20   VLAN0020                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
30   VLAN0030                          active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
40   VLAN0040                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
50   VLAN0050                          active    Fa0/19, Fa0/20, Fa0/21, Fa0/22
99   VLAN0099                          active    Fa0/10, Fa0/23, Fa0/24, Gig0/1
                                                 Gig0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Edificio 2, Sistemas (1er piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
20   VLAN0020                         active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
30   VLAN0030                         active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14
40   VLAN0040                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
50   VLAN0050                         active    Fa0/19, Fa0/20, Fa0/21, Fa0/22
99   VLAN0099                         active    Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

Edificio 2, Visitas (2do piso)

```
Switch>sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
20   VLAN0020                         active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
30   VLAN0030                         active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
40   VLAN0040                         active    Fa0/19, Fa0/20, Fa0/21, Fa0/22
50   VLAN0050                         active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14
99   VLAN0099                         active    Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>sh inter tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50
```

NAT

DINÁMICO

Guatemala

```
ip nat pool NAT_ADMIN 60.60.60.10 60.60.60.14 netmask 255.255.255.240
ip nat inside source list 10 pool NAT_ADMIN
ip nat inside source list 20 interface GigabitEthernet0/0 overload
ip classless
!
ip flow-export version 9
!
!
access-list 10 permit 174.10.0.0 0.0.0.127
access-list 20 permit 174.10.0.128 0.0.0.63
```

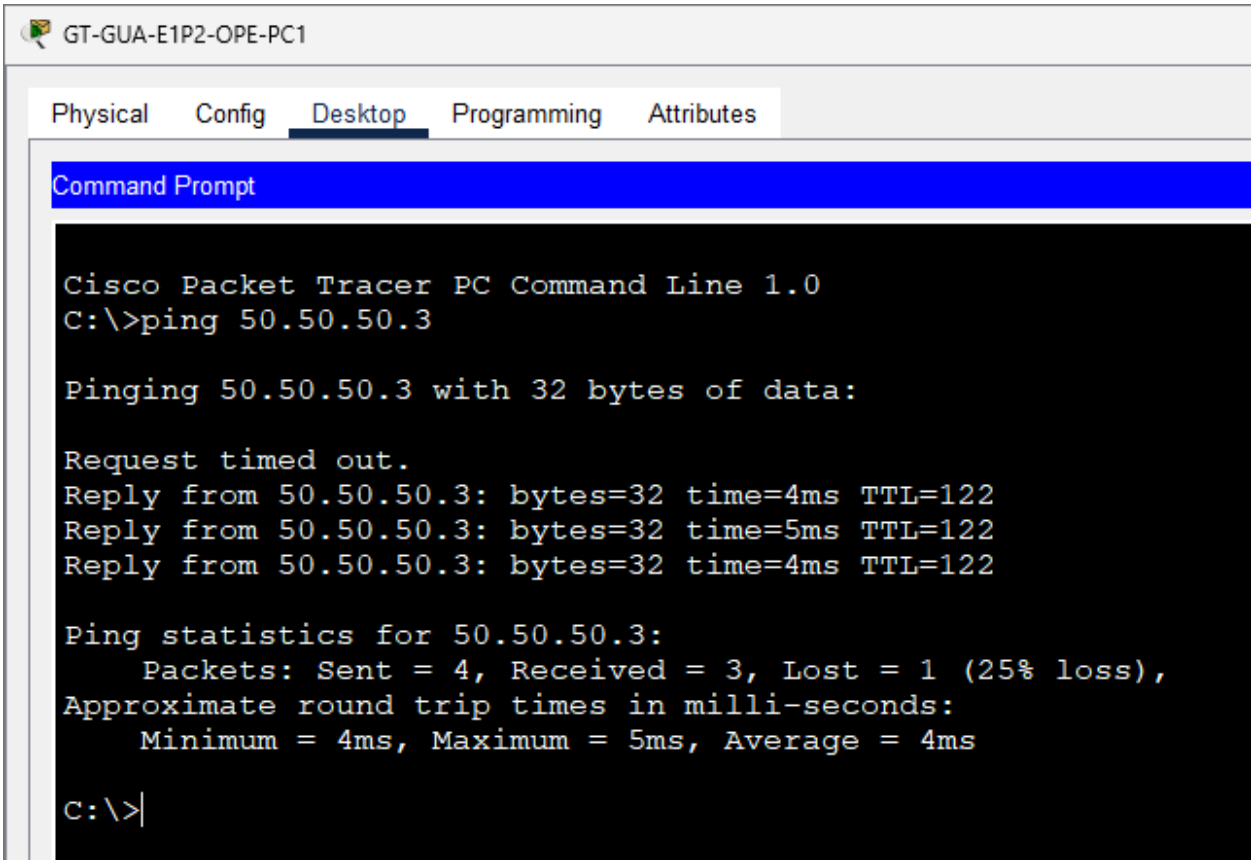
ESTÁTICO

Miami (Estados Unidos)

```
!
ip nat inside source static 175.168.20.66 50.50.50.3
ip classless
```

```
interface GigabitEthernet0/0
 ip address 50.50.50.2 255.255.255.240
 ip nat outside
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/1.40
 encapsulation dot1Q 40
 ip address 175.168.20.65 255.255.255.248
 ip nat inside
!
```

PAT

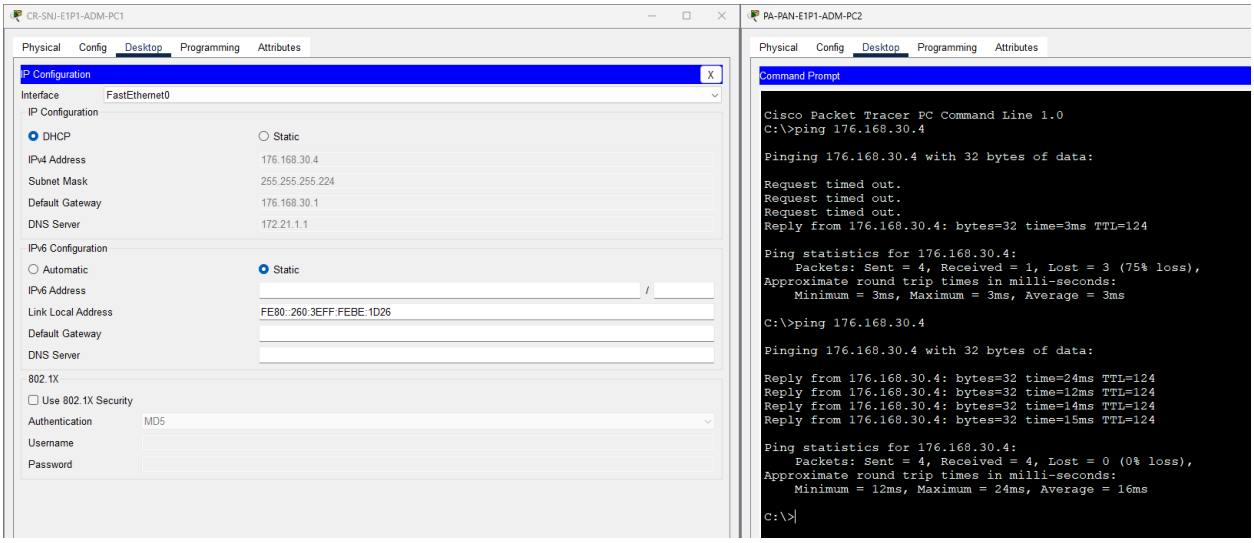


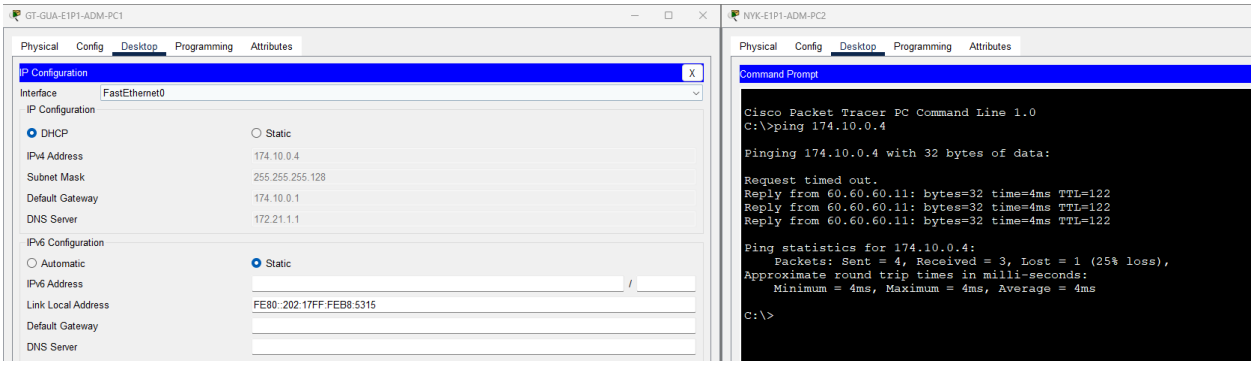

```
Router#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 50.50.50.3:2      175.168.20.66:2    60.60.60.2:2       60.60.60.2:2
icmp 50.50.50.3:3      175.168.20.66:3    60.60.60.2:3       60.60.60.2:3
icmp 50.50.50.3:4      175.168.20.66:4    60.60.60.2:4       60.60.60.2:4
---  50.50.50.3        175.168.20.66      ---                ---
```

PRUEBAS REALIZADAS

PING

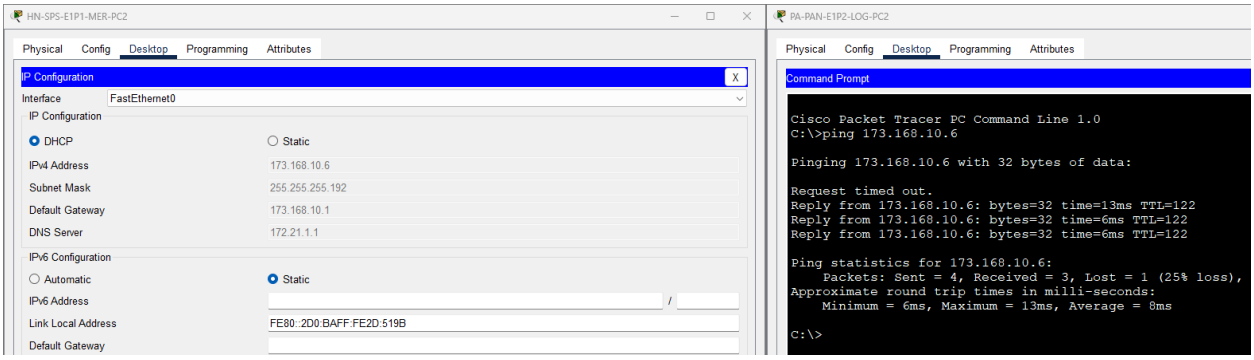
Entre ciudades e Inter VLAN

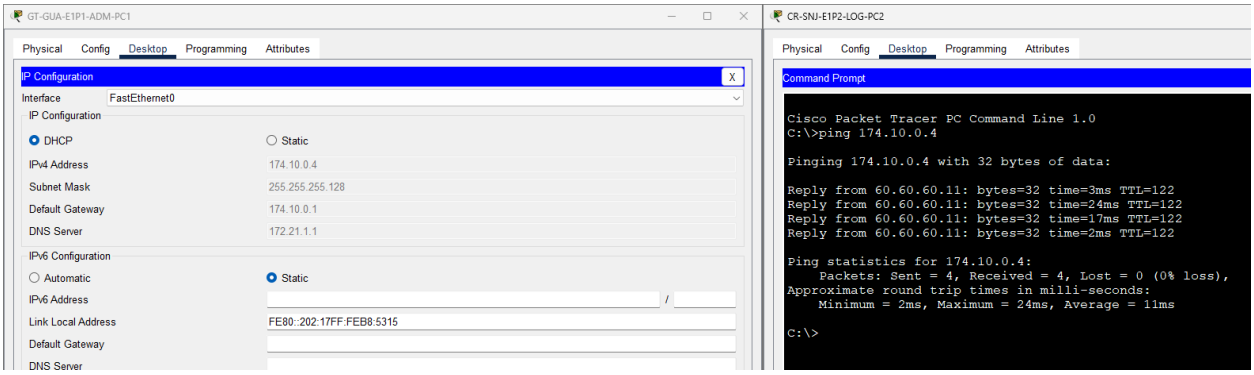




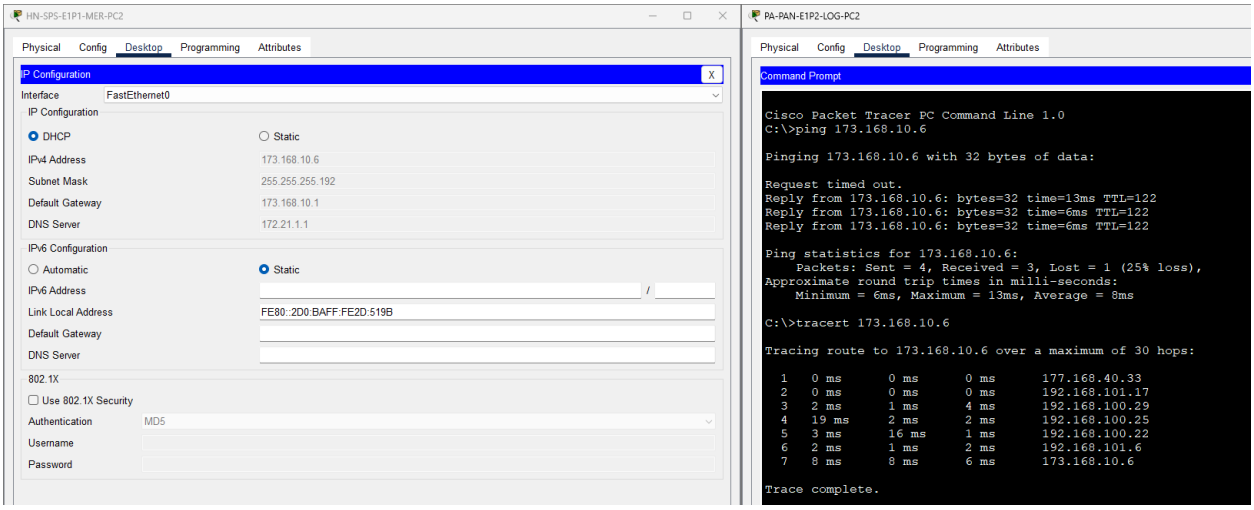
PING

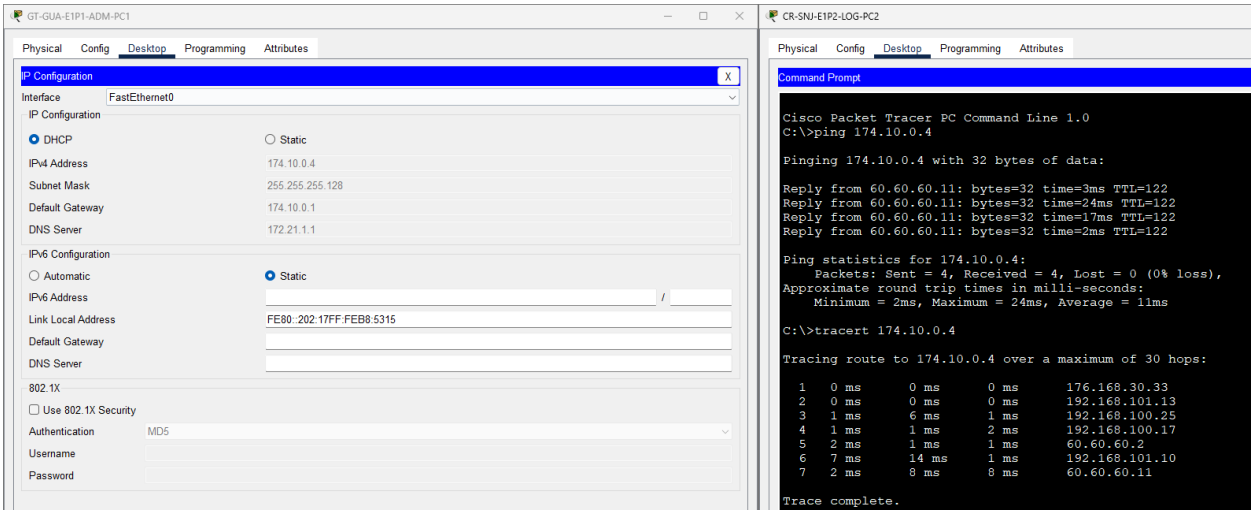
Entre ciudades e Inter VLAN distintas




GT-GUA-E1P1-ADM-PC1
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 174.10.0.4
Subnet Mask 255.255.255.128
Default Gateway 174.10.0.1
DNS Server 172.21.1.1
IPv6 Configuration
Automatic Static
IPv6 Address
Link Local Address FE80::202:17FF:FEB8:5315
Default Gateway
DNS Server
CR-SNJ-E1P2-LOG-PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 174.10.0.4
Pinging 174.10.0.4 with 32 bytes of data:
Reply from 60.60.60.11: bytes=32 time=3ms TTL=122
Reply from 60.60.60.11: bytes=32 time=24ms TTL=122
Reply from 60.60.60.11: bytes=32 time=17ms TTL=122
Reply from 60.60.60.11: bytes=32 time=2ms TTL=122
Ping statistics for 174.10.0.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 24ms, Average = 11ms
C:\>

TRACERT


HN-SPS-E1P1-MER-PC2
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 173.168.10.6
Subnet Mask 255.255.255.192
Default Gateway 173.168.10.1
DNS Server 172.21.1.1
IPv6 Configuration
Automatic Static
IPv6 Address
Link Local Address FE80::2D0:BAFF:FE2D:519B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
PA-PAN-E1P2-LOG-PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 173.168.10.6
Pinging 173.168.10.6 with 32 bytes of data:
Request timed out.
Reply from 173.168.10.6: bytes=32 time=13ms TTL=122
Reply from 173.168.10.6: bytes=32 time=6ms TTL=122
Reply from 173.168.10.6: bytes=32 time=6ms TTL=122
Ping statistics for 173.168.10.6:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 13ms, Average = 8ms
C:\>tracert 173.168.10.6
Tracing route to 173.168.10.6 over a maximum of 30 hops:
1 0 ms 0 ms 0 ms 177.168.40.33
2 0 ms 0 ms 0 ms 192.168.101.17
3 2 ms 1 ms 4 ms 192.168.100.29
4 19 ms 2 ms 2 ms 192.168.100.25
5 3 ms 16 ms 1 ms 192.168.100.22
6 2 ms 1 ms 2 ms 192.168.101.6
7 8 ms 8 ms 6 ms 173.168.10.6
Trace complete.


GT-GUA-E1P1-ADM-PC1
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 174.10.0.4
Subnet Mask 255.255.255.128
Default Gateway 174.10.0.1
DNS Server 172.21.1.1
IPv6 Configuration
Automatic Static
IPv6 Address
Link Local Address FE80::202:17FF:FEB8:5315
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
CR-SNJ-E1P2-LOG-PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 174.10.0.4
Pinging 174.10.0.4 with 32 bytes of data:
Reply from 60.60.60.11: bytes=32 time=3ms TTL=122
Reply from 60.60.60.11: bytes=32 time=24ms TTL=122
Reply from 60.60.60.11: bytes=32 time=17ms TTL=122
Reply from 60.60.60.11: bytes=32 time=2ms TTL=122
Ping statistics for 174.10.0.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 24ms, Average = 11ms
C:\>tracert 174.10.0.4
Tracing route to 174.10.0.4 over a maximum of 30 hops:
1 0 ms 0 ms 0 ms 176.168.30.33
2 0 ms 0 ms 0 ms 192.168.101.13
3 1 ms 6 ms 1 ms 192.168.100.25
4 1 ms 1 ms 2 ms 192.168.100.17
5 2 ms 1 ms 1 ms 60.60.60.2
6 7 ms 14 ms 1 ms 192.168.101.10
7 2 ms 8 ms 8 ms 60.60.60.11
Trace complete.